University of Toronto
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APS112
Conceptual Design Specifications (CDS)

Project #	181	Date	Mar. 22, 2026

Project Title	Automated Package Separator
Client	Sinai Health Pharmacy
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3499 words — State your word count (excludes Cover Page, Executive Summary, References and Appendices).

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Executive Summary

This document outlines the problem presented by the client, Sinai Health Pharmacy, who are seeking a design solution to automate the pill strip separation process. This request was made due to the occupational hazard that repetitively separating long strips causes and reveals a client's need for a technology that eliminates physical strain due to the absence of a reliable method to cut strips without injury risk.

This report documents the pharmaceutical packing facility at Mount Sinai Hospital. The service environment describes the physical area, including images of equipment, such as the BD ATP packer that creates long pill strips. The design only interacts with authorized staff and has no Wi-Fi connection, as that is outside of the scope with minimal influence on the design.

Interest holders play a crucial role in the design process, as many solutions will impact them explicitly and implicitly. Pharmacy technicians hold the highest influence and impact on the design, as they work with the prepackaged pills every day. This is followed by Sinai Health and the Biomedical Engineering Department, as their goals must align with the project's outcome.

The detailed requirements were generated through external research and analysis of the environment and were informed directly through client communication. The main function of this project is the separation of packaged pills into specific strip lengths. Design objectives were then derived to establish measurable goals for the project. The top objectives are to minimize human interaction while maximizing loading capacity and success rate. Finally, constraints discuss specific limitations of design solutions. The most important constraints include multiple cutting intervals, compatibility with pill sizes, and concealment of dangerous parts.

Designing a solution that aims to close the gap in pill separation stems directly from the project requirements. The solution space is filled with a wide range of means to separate pills, which go through an initial check and organization, then through a morph chart, which outputs full solutions. Feasibility checks and other analysis techniques are used to generate three final designs. Those three conceptual designs were narrowed down to PharmaSlice, ThermoCut, and SliceFlow, which use a variety of means to cut and move pill packages. Finally, the Pugh Method was used to contrast the final designs, in which the PharmaSlice demonstrated how it best met all the objectives, therefore filling the gap the most. Detailed testing specifications and a timeline is identified in the Measure of Success section for PharmaSlice.

This report covers the clients' gap and need and further analyzes the relevant scope and interest holders. It lays out the project requirements, which dictate the guidelines for creating a design solution that meets the purpose, goals, and limitations of the project. The design space is thoroughly explored through numerous tools and generated solutions, where the final chosen design, PharmaSlice, is found to meet the detailed requirements the best.

1.0 Introduction

Mount Sinai Hospital, located in downtown Toronto, delivers top-tier care to numerous patients and houses an in-hospital pharmacy. The pharmacy is crucial for patients and effective hospital workflow, as it produces single-packaged pills to be delivered to patients for 24-hour care. The client, Sinai Health Pharmacy, has identified an issue related to hand strain and the inefficiency of manually separating pill packages [1]. This report outlines design requirements by analyzing the client's need, environment, and interest holders. Based on the project's functions, objectives, and constraints, multiple design solutions are created and honed down using various analytical tools. This report will propose three design ideas and will identify the most suitable one.

2.0 Problem Statement

Pharmacies at Sinai Health System package oral medications into long unit-dose strips (Figure 1) using automated packaging machines, after which pharmacy staff must manually separate these strips into strips of 2, 5, or 10 pills for patients. This separation task relies heavily on repeated hand motions such as tearing, gripping, and counting, resulting in reported repetitive strain injuries from staff [2].



Figure 1. Unit-dose strips [1]

This results in a gap in the current workflow: no effective and reliable method exists to separate standardized unit-dose strips without exposing workers to the risk of injury. Consequently, there is a need for a technology that can perform this separation process efficiently while minimizing physical strain and maintaining medication package integrity and barcode (Figure 2) readability.

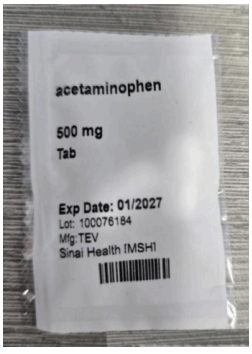


Figure 2. General package labelling [1]

The scope of this project is limited to designing a solution that operates within the Mount Sinai and Hennick Bridgepoint Hospital pharmacy environments, accommodating plastic unit-dose packs of a single standardized dimension and using standard electrical outlets. This scope is constrained to ensure feasibility, compatibility with existing processes, and risk reduction. The intended end users of the design are pharmacy technicians who regularly perform medication packaging and preparation tasks.

3.0 Service Environment

This section outlines the environment in which the solution will operate. Specifically, for use in Mount Sinai and Hennick Bridgepoint Hospitals.

3.1 Physical Environment

The design will operate indoors within the pharmacies of Mount Sinai and Hennick Bridgepoint Hospitals (Figure 3). The Mount Sinai pharmacy spans 7000 square feet, and the floor is coated with sealed 6mm poly [3]. The pharmacies both include large, permanent machinery, including the Automatic Tablet Packager (ATP) machine from TCGRx [1] and the PillPick Unit Dose System, as seen in Figures 4 and 5. As an acute-care hospital [4], high compliance with Public Health Ontario pharmacy CRMD guidelines is expected, including sterile prep areas, localized air-exchange systems, and workstations with HEPA laminar flow systems [5]. Key environmental factors are highlighted in Table 1.

Table 1. Table of environmental factors

Aspect of the environment	Data
Temperature	20-25°C [6]
Humidity	≤60% [6]
Electrical Supply	15A/125V [7]
Sound Levels	<85 dB [8]



Figure 3. Mount Sinai Hospital Pharmacy [3]



Figure 4. ATP machine (TCGRx) [1]



Figure 5. PillPick Unit Dose System at Mount Sinai [9]

3.2 Living Environment

This design will be used in areas of the hospital that are only accessible to authorized hospital staff, meaning patients and the general public will not be able to access this environment (Sinai Health Pharmacy, private communication, Feb. 2, 2026). The primary living things of interest within this environment would be microbes. The area in which the solution would operate will

have no HVAC engineering or other procedures to control microbial presence (Sinai Health Pharmacy, private communication, Feb. 2, 2026).

3.3 Virtual Environment

The virtual environment includes secure hospital networks that manage patient data, medical histories, and treatment plans through the Electronic Health Record system. Examples include the patient-facing MyChart portal, employee remote-access environment Citrix, and Novari ATC surgical waitlist system [10][11][12]. Cellular network and hospital Wi-Fi infrastructure are also present throughout the hospital.

4.0 Interest Holders

The interest holders identified in Table 2, derived from an Impact-Influence Matrix (Appendix A and Appendix B), have varying levels of influence. Their expectations and concerns help shape the project's detailed requirements.

Table 2. Interest Holders Ordered from Highest to Lowest Relevance (Appendix B)

Interest Holder	Impact	Influence
Pharmacy Technicians	● As primary users of the design, the solution has a high impact on technicians' daily workflows and patient care. ● The speed, consistency, and reliability of the design directly affect the ability to administer medication safely and on time.	● The technicians provide feedback on the design and can assess it. ● A design that damages packages, jams frequently, or requires constant maintenance would influence levels of productivity and increase technician workload. [1]
Sinai Health	● This design directly contributes to Sinai Health fulfilling their vision of becoming "Canada's leading integrated health system," which relies on innovations that improve current healthcare delivery systems and staff safety [13]. ● The project helps them achieve this goal by addressing repetitive strain injuries during medicine	● Sinai Health System's priorities highly influence both the hospital's operations and this project's feasibility and objectives [16]. ● We need to align our project with their long-term vision for "outstanding quality, safety, and patient experience" and "efficiency in care" [14]. ● As such, their requirements shape the project's scope and

	packaging, which directly supports their value of "compassion" towards healthcare workers [14].	constraints, ensuring the final design positively contributes to the system's goals. [14].
Mount Sinai Biomedical Engineering Department [10]	• Their high impact comes from their responsibility for the entire lifecycle, from validation of technical standards to installation to maintenance and repair of over 13,000 medical devices across the hospital. • This involves "scheduling service visits and collaborating with manufacturers" [15].	• Their expertise in machinery indicates strong influence over the design's serviceability and compatibility. • With a team resolving over 8,000 support calls annually, their requirement for a reliable, easily maintainable, and safe design is an important project constraint [15].

5.0 Detailed Requirements

The following section details the specific functions, objectives, and constraints for the project.

5.1 Functions

Table 3 below describes the functions of the design, chosen using the Black Box Method (Appendix C) and the Functional Basis (Appendix D).

Table 3. Design Functions

Category	Function	
Primary	PF. 1	Separate tablet packages
Secondary	SF. 1	Advance the tablet strip within the machine
	SF. 2	Reposition tablets within the package pouch
	SF. 3	Output tablet strips

5.2 Objectives

Table 4 describes the objectives, derived from a how-why chart and ranked by importance, chosen using a pairwise comparison (Appendix E).

Table 4. Design Objectives (Appendix E)

	Objective	Metric	Goal	Justification
O. 1	Minimize Human Interaction	Minutes spent interacting with the machine (min)	<60 min per day	Per client's specifications, pharmacy technician's interaction with the design should be limited to the feeding of strips into the machine while checking for damaged pills (Sinai Health Pharmacy, private communication, Feb. 2, 2026).
O.2	Maximize total production	Number of strips cut per hour (#)	≥1750 strips per hour	The design should not be a limiting factor in the pharmacy workflow. With a maximum 8000 pouches/day, it would take under five hours to complete. (Sinai Health Pharmacy, private communication, Feb. 2, 2026).
O. 3	Maximize success rate	Undamaged package per 1000 packages (#)	Success rate ≥99.9%	Based on client expectations for reliability. Failures, including jammed/damaged packages would increase staff workload [1][17][18]. $Success\ rate = 100\% * \left\lvert \frac{total - failed}{total} \right\rvert$
O. 4	Maximize loading capacity	Roll capacity	≥500 packs	The ATP machine outputs a maximum of 500 packets in a roll. Increasing loading capacity reduces the need for staff intervention, increasing efficiency in order to achieve production quotas [1].
O. 5	Minimize maintenance frequency	Frequency of component swap (#)	≥3 months	Sinai Health has only 17 biomedical engineers handling over 500 service calls monthly. Therefore, minimizing maintenance frequency is paramount to reduce biomedical staff workload [19] [16].
O. 6	Maximize internal visibility from outside	Area for visual access (cm^2)	Length ≥180mm (3x pouch) Width ≥40mm (1x pouch)	Enables pharmacy staff to supervise the machine and intervene in-case of failures. Dimensions accommodate visibility for 3 pouches (length) and one pouch width.
O. 7	Minimize dimensions	Length (cm) Width (cm)	≤80.0cm ≤62.0cm	Based on the average dimensions of office and pharmacy equipment,

		Height (cm)	≤116.9cm	ensuring device compatibility with service environment [20] [21].
O. 8	Minimize spending	$ CAD	≤$10000	The client stated the budget of $10000. (Sinai Health Pharmacy, private communication, Feb. 2, 2026)

5.3 Constraints

Table 5 lists the constraints that describe what the design solution must be, directly influenced by the client statement, service environment, and other subsections of this report.

Table 5. Constraints Ordered from Highest to Lowest Importance (Appendix F)

	Constraint	Metric	Limit	Justification (Sinai Health Pharmacy, private communication, Feb. 2, 2026).
C. 1	Must accommodate different cutting intervals	Number of individual pill packages per cut strip (#)	2, 5, 10	Described as a necessity on slide 6 of the client presentation [1].
C. 2	Must have high cutting accuracy	Cutting error (mm)	± 3mm	Each pouch contains a seam that must be cut within a 3 mm tolerance to prevent contamination. (Figure 6)
C. 3	Must not have exposed sharp moving parts	Length of exposed moving part (cm)	0cm	The client expressed that any exposed sharp parts will put staff at risk and is therefore unviable. [1]
C. 4	Must be compatible with different pill sizes	Range of sizes of pills that are accepted (mm)	6-22mm	The largest and smallest pills the machine must accommodate are, levothyroxine and calcium carbonate, respectively [22][23]. Pills are measured based on their longest axis in mm.
C. 5	Must have reduced implementation complexity	Number of external database connections needed (#)	0 databases	The client said that any design requiring software or a connection to an external database is outside the scope of this project, as it would significantly increase costs.



Figure 6. Cutting area on packages.

6.0 Generation, Selection and Description of Alternative Designs

The team first individually generated solution means, which were later jointly developed into full conceptual designs, all achieving the functions of the design. Solutions were compared against objectives and constraints and narrowed down. The process is shown in Figure 7 and demonstrated in Appendix G.

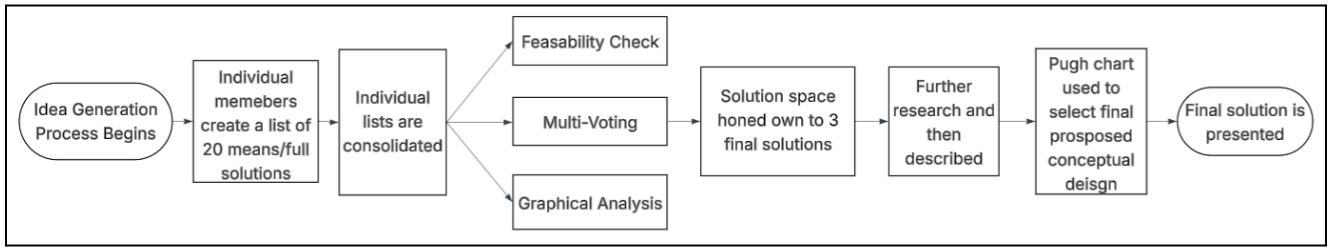


Figure 7. Idea selection to final design process flowchart

6.1 Idea Generation Process

Idea generation began with individual brainstorming, blue-sky thinking, and SCAMPER methods. Team members then presented their means, which were consolidated. After removing duplicate ideas, 83 solution means remained (Appendix G.3). The list was categorized into design functions, and a morph chart transformed these to create full solutions (Appendix G.4)

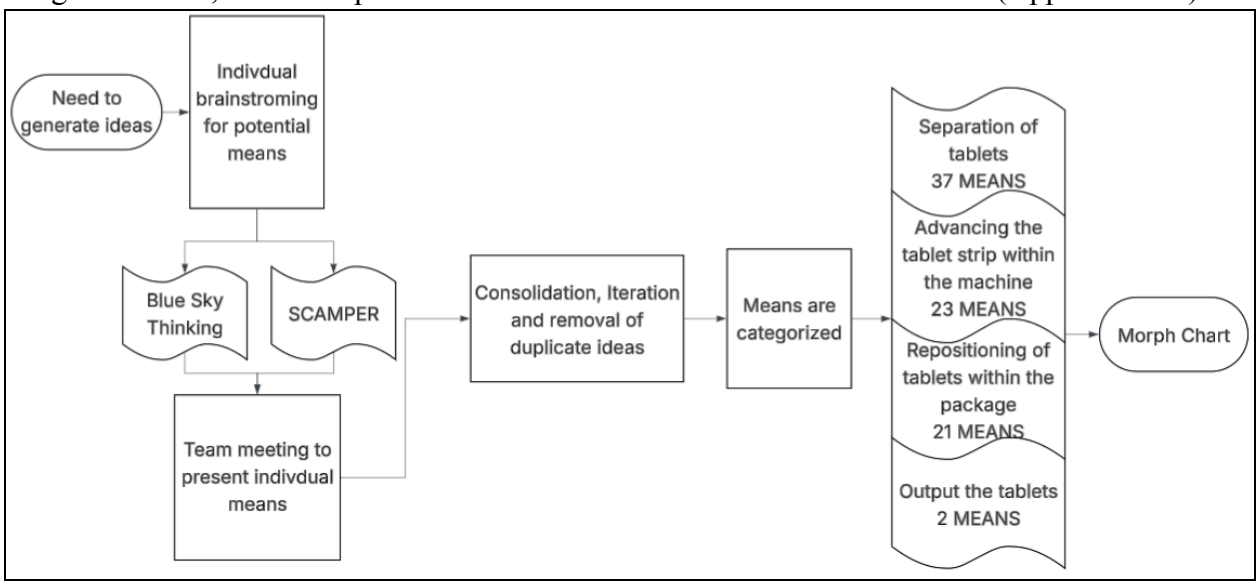


Figure 8. Idea Generation Process.

6.2 Alternative Design Selection Process

The design selection process outlined in Figure 9 narrowed 69 initial ideas down to three final options.

- **Consolidation (Appendix H.1):** Combining redundant and overlapping ideas refined the initial 69 ideas to 42.
- **Feasibility Check (Appendix H.3):** Eliminating ideas inconsistent with the project's requirements further reduced the pool to 21 viable solutions.
- **Multivoting (Appendix H.4):** Members voted to identify concepts that best support the project's main goals, narrowing the list to a further nine designs
- **Graphical decision chart (Appendix H.5):** The remaining concepts were evaluated against the two highest-ranked objectives, resulting in the top 3 solutions
- **Pugh Chart (Appendix I):** The top conceptual designs were compared to select the solution that best satisfied the project's objectives.

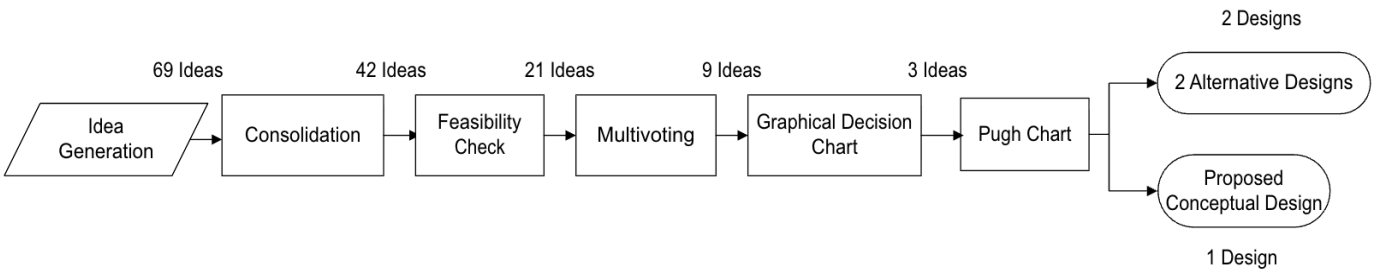


Figure 9. Idea Selection Process.

6.3 Alternative Design Descriptions

The following sections outline the three alternative designs.

6.3.1 Design 1 – PharmaSlice

The PharmaSlice design, noted in Figure 10, features a rotary paper trimmer mechanism mounted on a wooden rail, with wheel-driven advancement of the medication strip along a guided path. Two DC gear motors controlled by an Arduino Uno drive the trimmer cart and wheels, while positioning rails guide each package during cutting to ensure straight, vertical cuts.

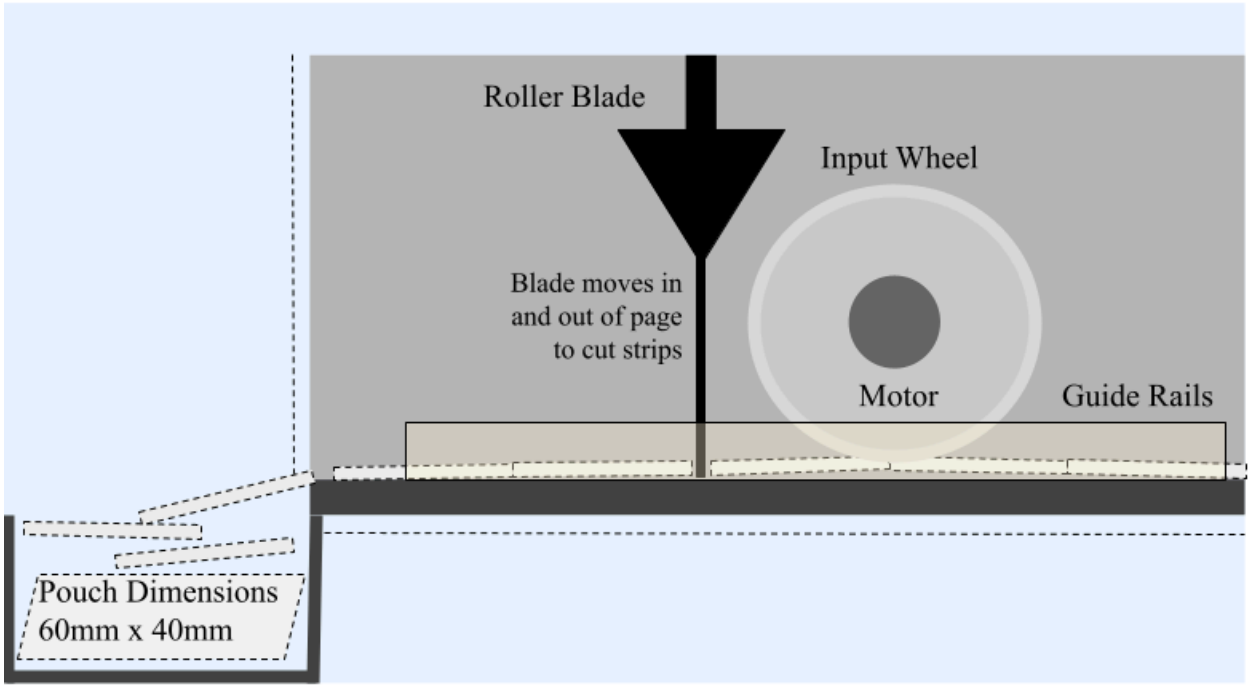


Figure 10. PharmaSlice (Side View).

Table 6: Design 1 Objectives Analysis, justification in Appendix J.1

Objective	Analysis
Human Interaction	The operator only loads strips and removes separated packages; automated advancement and cutting eliminate the need for manual tearing.
Total Production	The dual-motor system, with the output geared faster than the input, allows for continuous, jam-free operations with Arduino-controlled intervals. Projected to meet ≥1750 strips/hour.
Success Rate	Guide rails and friction from wheels ensure consistent cuts. Cutting timing can be tuned with the Arduino, projected to meet ≥99.0% success rate.
Loading Capacity	The open feed path accommodates continuous strips with no physical limit, projected to meet ≥500 pack requirements.
Maintenance Frequency	Off-the-shelf components are easily replaceable; the blade may require periodic replacement every 3 months.
Internal Visibility	Open-frame design allows for full visibility of operations, fulfilling the objective.

Dimensions	Compact footprint, fitting within 305x305mm, fits within objective limits and standard countertops.
Cost	The total prototype will cost under $200 CAD, if commercialized, it is projected to be well under $10000 CAD.

6.3.2 Design 2 – ThermoCut

ThermoCut, as shown in Figures 11 and 12, features a compact roller-driven device with a heated nichrome wire cutting mechanism. Medication strips are guided by rails and transported by DC motor-driven wheels to a fixed cutting location, where a heated 18 AWG nichrome wire slices through the PET package connections at above their melting point for clean, fray-free cuts [24]. After cutting, the pouches fall into a collection basket below. Traces of vaporized fumes are absorbed by a filter.

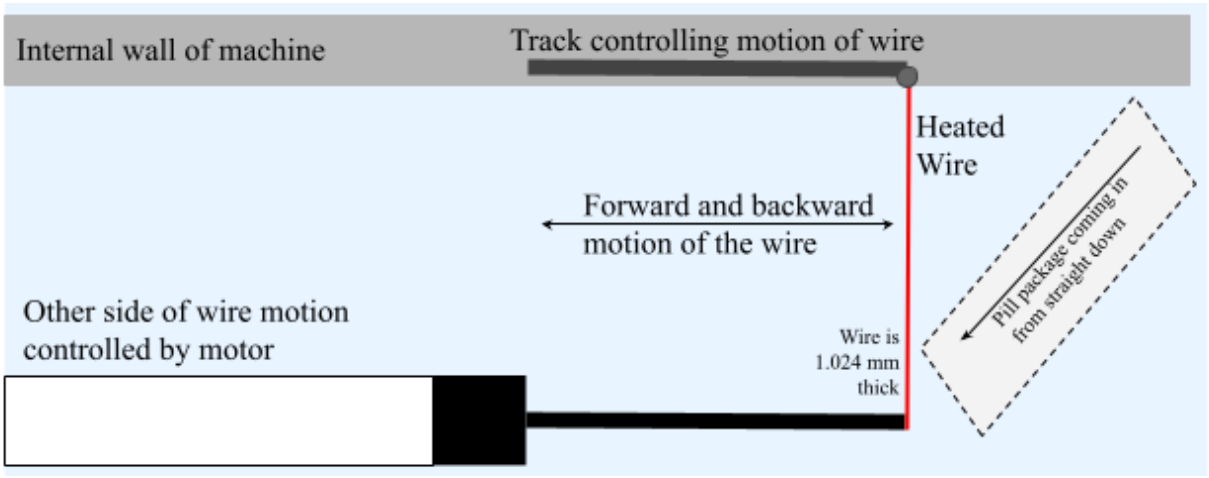


Figure 11. ThermoCut (Top View)

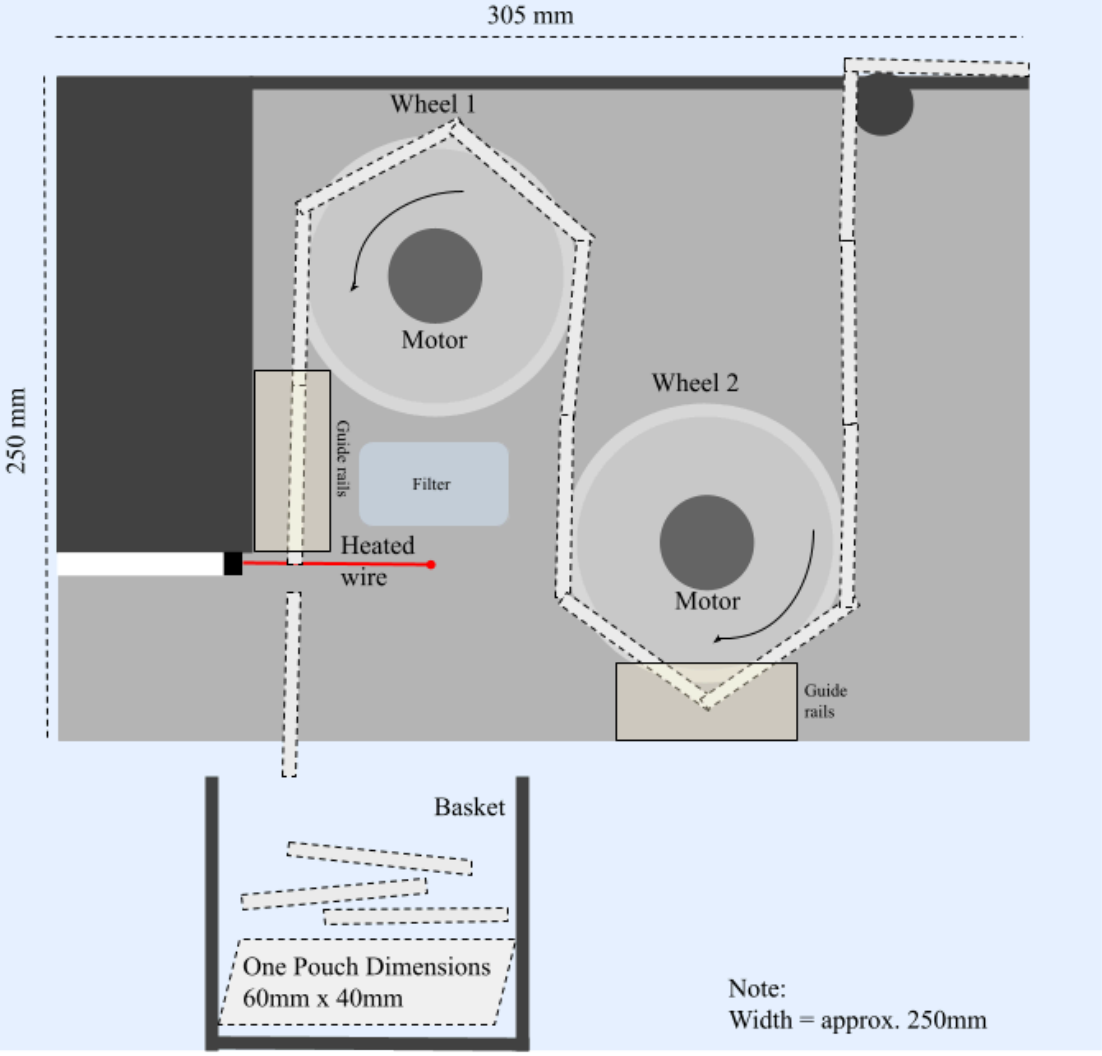


Figure 12. ThermoCut (Side View).

Table 7: Design 2 Objectives Analysis, justification in Appendix J.2

Objective	Analysis
Human Interaction	Rollers transport packs with minimal operator intervention. However, manually feeding and adjusting strips between the inner wheels are required for each reload.
Total Production	Motors regulate roller speeds, synced with wire cutting speed. Benchmarked heated wire cutters perform best at speeds of 400mm/min for thicker, 1cm styrofoams. Using this as a safe estimate, efficiency is projected to be ~1788 strips/hour [25].
Success Rate	Guide rails keep strips on path. The wire will be heated to 300°C, greatly above PET's (Sinai Health Pharmacy, private communication, Mar. 6, 2026) melting point of ~250°C, ensuring ≥99.9% success cuts without

	fraying [26].
Loading Capacity	Once aligned and loaded by staff, the design can function for an indefinite time, limited only by the input strip length of 500 (from ATP machine output).
Maintenance Frequency	Nichrome's max operating temperature of 1050°C is well above the service temperature required, contributing to wire durability [27]. Filter replacement will be varied but should be replaced every 3-12 months [28].
Internal Visibility	A 130 x 180mm transparent glass panel is mounted on the side to allow inspection of operations and jams.
Dimensions	Compact integrated design within 305 x 250mm, fitting within the standard of constraint and objective.
Cost	This design minimizes wheels used in aligning, while required materials, like nichrome wires and small motors, are low-cost and accessible. This allows the total system cost to remain under $10000 CAD.

6.3.3 Design 3 – SliceFlow

The SliceFlow design (Figure 13) features a motor-controlled [29] swivel blade mounted on a pivot joint [30] that rotates to cut along the perforated seam. A dual-conveyor belt system feeds the medication strip through the machine, with a gap between the belts to allow the blade to pass. Before cutting, a foam roller gently pushes pills to one side of the package, preventing blade contact, while a cleated slider bed [31] maintains package alignment during advancement. The entire machine is enclosed in transparent acrylic plastic.

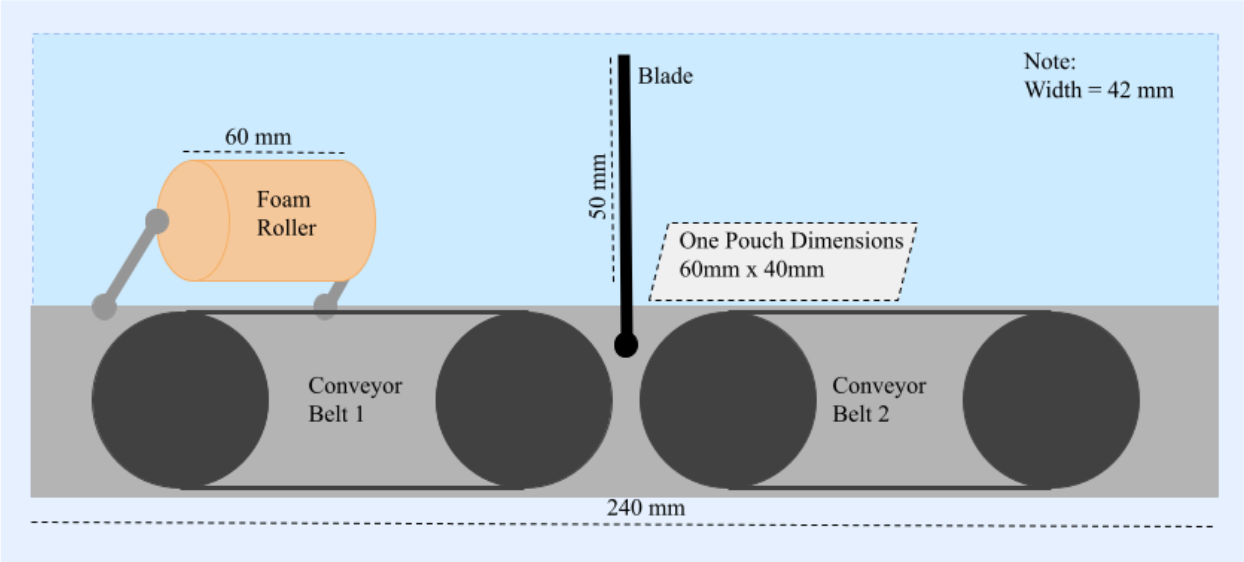


Figure 13. Side view of SliceFlow.

Table 8: Design 3 Objectives Analysis, justification in Appendix J.3

Objective	Analysis
Human Interaction	The only human interaction required with SliceFlow will be inserting new rolls of pouches. This total time will be below the 60 min/day goal.
Total Production	The dual-conveyor system enables continuous feeding and cutting cycles, with conveyor speeds and blade action times projected to meet ≥1750 strips/hour for 2-package batches.
Success Rate	Foam roller repositions pills away from the cut line, while the cleated slider bed maintains alignment, preventing failed cuts while meeting ≥99% success rate goal.
Loading Capacity	Given the open feeding system, this design has a very high loading capacity with no physical limitations.
Maintenance Frequency	Use of a carbide blade on thin, soft plastic pouches prevents rapid blade degradation, while the vertical cutting orientation minimizes edge wear.
Internal Visibility	The outside shell of the design uses transparent acrylic plastic, increasing the visibility much greater than the desired window.
Dimensions	SliceFlow is expected to fit within 42x240mm, fitting within the objective requirements of 800x620mm.
Cost	This design requires minimal motorized parts. Estimated to be within the CAD $10000 budget.

6.4 Proposed Conceptual Design Specifications

Through the use of a Pugh Chart and associated analysis (Appendix I), the top three conceptual designs were compared to analyze which one most effectively met the project's objectives. Of the designs outlined in section 6.3, the design we are recommending is Design 1 (PharmaSlice).

The client's original need was for "an effective and reliable method that exists to separate standardized unit-dose strips without exposing workers to the risk of injury." PharmaSlice fulfills this need by offering a design that provides an automated way to separate packages while meeting all objectives and constraints (including limiting human interaction). This will allow pharmacy technicians to separate up to 8,000 pouches of medicine each day, without causing overuse injuries and minimizing the use of human resources.

7.0 Measure of Success

This section outlines the bench-top test plan developed to validate that the proposed rotary paper trimmer design (PharmaSlice) meets the topmost objectives. Using a functional prototype with wheel-driven strip advance and positioning rails, the team will collect primary evidence on

success rate, production output, and required human interactions, three interdependent objectives ranked highest in throughput pairwise comparison and client feedback (Sinai Health Pharmacy, private communication, Feb. 2, 2026). Simultaneous testing is essential because failure in any objective listed above compromises the others. Misaligned cuts increase intervention time and reduce production output; feeding errors create waste and increase operator workload; and mechanical jams stop production entirely while risking package damage.

The test method follows established bench-top validation methods common in medical and manufacturing device development, where key mechanisms and functions are tested under controlled conditions [32][33]. Testing will be conducted by two team members, an operator running the machine/tests and a recorder focused on recording video data to ensure consistent testing [34]. The prototype will be assembled at Myhal MyFab using the following materials:

- **Wooden frame**
- **Two 60 RPM DC gear motors**
- **Arduino Uno**
- **Rail- mounted rotary paper trimmer**
- **~200 authentic hospital strips and office paper** (~110µm): a material matching the thickness and properties of LDPE/PET packaging used in the medicine strips (~110µm total thickness, verified per ASTM D6988), consistent with Health Canada GUI-0029 methods for using similar materials during testing when authentic samples are limited [35].

The testing procedure will occur over 3 weeks following the CDS deadline (Mar 22nd, 2026). This process can be seen in Figure 14.

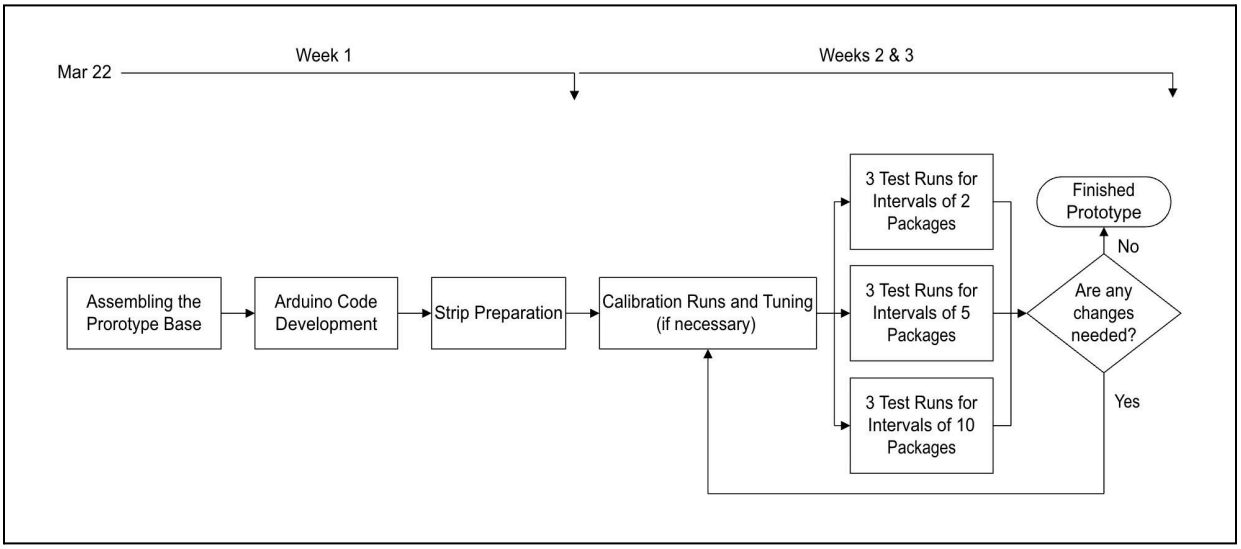


Figure 13. The Testing Steps and Timeline.

Data collection includes loading and intervention times recorded via video, total run time, packages processed, and a post-test visual inspection to categorize failures (misaligned cuts, torn packages, or jams). Success will be measured against 3 criteria: total daily human interaction under 60 minutes (extrapolated to 8000 packages), a production rate at or above 1750 strips/hour, and a success rate of at least 99.9% undamaged packages.

Due to testing under controlled environments, the test carries several limitations: only 200 authentic samples are available, requiring substitute materials; visual inspection may miss minor damage; the controlled testing environment may not reflect hospital workflow interruptions; and nine runs provide a limited sample size.

Several assumptions were made to simplify the testing process. Namely, the paper thickness simulates the actual packaging, results from 1000 test packages can be extrapolated to a full day of production, the two-person testing team improves reliability, and visual inspection is sufficient to detect integrity-compromising failures. If the prototype narrowly misses one success criterion while exceeding in others, the team will document the gap and note potential future modifications.

8.0 Conclusion

This report outlines the design requirements, as well as potential solutions for the automation of strip separation for packaged pills. The problem statement specifies the need for an effective solution that will support the needs of in-hospital pharmacies. Important interest holders are evaluated for their impact and influence on future designs, such as pharmacy technicians. Client needs are further categorized in sections, functions, objectives, and constraints. Following these detailed requirements, many design solutions were generated and evaluated, with attention to objective analysis to hone the solution space down to three designs: PharmaSlice, ThermoCut, and SliceFlow. Ultimately, PharmaSlice was selected as the proposed design through Pugh Chart analysis, after which a detailed testing timeline was developed.

9.0 References

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Appendix A: Interest Holders Brainstorm

Table A1 shows the list of all interest holders identified through an Impact-Influence Matrix (Appendix B). This matrix helped categorize stakeholders based on how much they affect the system and how much the system affects them. The resulting list provides a comprehensive view of everyone with a meaningful role in or connection to the dispensing system.

Table A1. List of all interest holders.

#	Interest Holder
1	**Pharmacy Technicians**
2	Pharmacy Managers
3	**Biomedical Engineering Department**
4	Ergonomic Specialists
5	Canadian Union of Public Employees (CUPE)
6	BD/ Parata (ATP Vendors)
7	Procurement / purchasing department
8	Institute for Safe Medication Practices (ISMP)
9	Toronto General/ Sick Kids/ Princess Margaret Hospitals
10	Supply chain and inventory management staff
11	Patients
12	Cleaning Staff
13	Facility maintenance staff
14	Pharmaceutical Drugs Directorate (PDD)
15	Medical Devices Directorate (MDD)
16	Health Products and Food Branch (HPFB)
17	Assurance & Compliance Office
18	**Sinai Health**
19	Healthcare Insurance Providers
20	Medical Device Reprocessing (MDR) Technicians

Appendix B: Interest Holders Ranking

Table B1 ranks the interest holders from Appendix A using an Impact-Influence Matrix to evaluate each group's role in the project. Interest holders were positioned based on how strongly the system affects them and how much authority they have to shape its design and operation. The resulting categories clarify which groups must be prioritized during decision-making and which require ongoing consideration throughout the project.

Table B1. Analysis of the interest holders from Table A1.

High Impact, Low Influence 11, 12, 13	**High Impact, High Influence** 1, 3, 7, 18
Patients (11), Cleaning staff (12), and	Pharmacy technicians (1), Biomedical

Facility maintenance staff (13) are placed here because while they are significantly impacted by the device's operations, they have little formal authority to influence its design and implementation. Patients are the beneficiaries of the design, as the design's reliability directly influences medication safety and delivery speeds. With the design planned across multiple wings, cleaning staff significantly impacted by the design's ease to clean and procedures. Similarly, the maintenance staff are responsible for the design's upkeep, where a poorly serviceable design could increase workload and decrease operational uptime. These interest holders share a high impact from the design due to their daily interactions, and their low influence results from not being included directly in the design process.	engineering department (2), Pharmacy technicians (7), Sinai Health (18) each are essential to shaping the final design with their opinions. Biomedical engineers are required to maintain and upkeep the design, ensuring it is safely operational. Pharmacy technicians are the primary users of the device and benefit directly from the improvement in workflow the machine provides. Lastly, Sinai Health defines the project's alignment with their mission statement and organizes the possible large-scale distribution of the machine across its hospitals.
Low Impact, Low Influence 4, 5, 10, 15, 19, 20	**Low Impact, High Influence** 2, 6, 8, 9, 14, 16, 17
Ergonomic specialists (4), CUPE (5), Supply chain staff (10), MDD (15), Insurance providers (19), and MDR technicians (20), have limited influence over the implementation or design process. They act as advisors, upholding their regulations and protocols, however, are not specific to this design, and in most cases, do not apply to our scope. Their needs only become relevant in later stages, such as commercialization or implementation with other medical machines, resulting in these interest holders in this quadrant for the current project phase.	Pharmacy manager (2), BD/ Parata (6), ISMP (8), other hospitals (9), and Regulatory/ compliance offices (14, 16, 17) hold significant influence to enable or constrain the project but are less affected by it in the daily operations. The vendor (6) controls the compatibility with existing packages and ATP systems. ISMP (8) and other regulatory groups set regulations around the design. Nearby hospitals (9) represent the similar workflows the design will have to handle if it is to be adopted in the future, influencing scalability. The various internal offices (14, 16, 17) ensure compliance with Sinai Health policy compliance. While the daily operations of the machine do not depend on this device, their standards and relations set significant bounds on the design, giving them strong influence despite little direct impact.

Appendix C: The Black Box Method

This appendix shows the system used to define the design's functions, the Black Box Method (Table C1.). By focusing only on observable inputs and outputs, the method clarifies what the machine must achieve without specifying internal mechanisms. This approach helps establish the essential functional requirements that guide design work.

Table C1. Black box method to create functions

Input		Output
Mass: - Strips of pills enter the machine		Mass: - Individual packages of pills (strips of 2s, 5s, 10s)
Information: - Strips of pills with information on barcode goes into the machine		Information: - Strips of pills with information on barcode comes out of the machine (no change)
Energy: - N/A		Energy: - N/A

Appendix D: The Functional Basis

The functional basis identifies the core function of the system using a specific framework. For this project, the functional basis for this project would be Separate Mass, reflecting the device's role in dividing continuous pill strips into smaller packaged units. This classification established the foundation for defining the system's mechanical and operational requirements.

Appendix E: Objectives

This appendix outlines the design objectives generated in Table E1, which shows the full brainstorming list. These objectives were then prioritized through pairwise comparison shown in Table E2, establishing their relative importance for guiding the system's development.

Table E1. Objective Brainstorming List

Objectives Brainstorm
Does not require software updates (complexity)
Less maintenance (money cost and amount of time per month)
Can load an entire strip per feed (e.g., >=100 packs per feed)
Compact in size (Measured in m^3)

Ideally works pill-size dependent Should be transparent to allow visual inspection Should ideally use a tearing motion Should work continuously for 2-5 hours

Table E2. Pairwise Comparison of the ObjectivesAppendix F: Constraints

	Success Rate	Cost	Human Interaction	Loading capacity	Allow visual inspection	Require minimal maintenance	Dimensions	Total Production	**Score**
Success Rate	X	1	0	1	1	1	1	0	**5**
Cost	0	X	0	0	0	0	0	0	**0**
Human Interaction	1	1	X	1	1	1	1	1	**7**
Loading Capacity	0	1	0	X	1	1	1	0	**4**
Allow visual inspection	1	0	0	0	X	0	1	0	**2**
Require minimal maintenance	0	1	0	0	1	X	1	0	**3**
Dimensions	0	1	0	0	0	0	X	0	**1**
Total Production	1	1	0	1	1	1	1	X	**6**

This appendix lists the design constraints identified during the project, as summarized in Table F1, which captures the full constraint brainstorming output. These constraints define non-negotiable requirements the design must meet to ensure safety, functionality, and compatibility with existing workflows.

Table F1. Group Constraint List

Constraint Brainstorm
No blade/ exposed blade usage (Proly Constraint) From ATP machine able to separate into strips of 1, 2, 5, 10 (constraint) Eliminate tearing motion required by staff

Handle different pill sizes Temperature change <= few degrees Emergency stop function Error rate (1 in 1000 at least)

Appendix G: Idea Generation Process

Appendix G documents the full development of means used to generate design concepts. Section G1 contains each team member's individual brainstorming, created using timed-thinking, blue-sky thinking, benchmarking, and in some cases early morph chart based solutions. Section G2 presents the combined list of means organized by function, followed by Section G3, which consolidates these entries and highlights duplicates in blue. Finally, Section G4 includes the morph chart constructed from the consolidated list, which was used to generate the team's set of 69 design ideas.

G.1: Individual Brainstorming

Name	Brainstorming
Alex	1. Robotic arms that tear bags based on a vision system 2. Abrasive discs and a roller for transporting packages 3. Regular guillotine and tracks: as the packages move through, a guillotine cuts connections 4. Have two clamps that grip the bags and tear, paired with a vision system 5. Any solution that uses low temperature to make the plastic bags brittle to assist in separation 6. Still track/conveyor belt, but have saw-tooth blades for clean separation (either punch or slice back and forth) 7. Use a roller conveyor and highly pressurized water 8. Heated knife that more easily cuts through bags. (potentially wire, low melting point of plastics, need a specific type or Tm determined) 9. Ultrasonic blades? 10. Conveyor belt and high-energy laser 11. Torque-driven designs that twists to separate
Alba	Separation Mechanics: 1. Saw 2. X-acto Knife 3. Laser Cutter 4. Punching/Stamping 5. Waterjets

6. Hot Knife
7. Freezing (causing brittleness)
8. Robot Arm

Movement:

1. Robotic Arm
2. Carry by Hand
3. Use gravity to unroll (on a slant)
4. Conveyor Belt
5. Rollers
6. Robot Car
7. Vacuum
8. Floating on Water

Keeping in Place:

1. Clamps (on edges of strips)
2. Suction
3. BluTack
4. Robotic Arm
5. Carry by Hand
6. Rollers (on top)
7. Robot Car
8. Vacuum
9. Floated on Water
10. Metal Plates (on edges)

Morph Chart:

	Means	Means	Means	Means	Means	Means
Separate tablet packages	Saw	X-acto Knife	Punching /Stamping	Robot Arm	Laser Cutter	Hot Knife
Advance the tablet strip within the machine	Robot Arm	Gravity	Conveyor belt	Rollers	Vacuum	

Boen	Sensing & Alignment 1. Color / Contrast Sensor - Detects the printed line or varying light transmission between the pouch and the pill to trigger the cutting mechanism. 2. Mechanical Micro-switch - Uses a small roller arm that physically drops into the perforated groove between pouches to activate the cut. 3. Stepper Motor Dead-Reckoning - Feeds the strip by a precise, pre-programmed distance (e.g., 50mm per pouch) without relying on external optical sensors. 4. V-shaped Guide Funnel - A purely passive mechanical guide that flattens and aligns the twisted strip before it enters the cutting zone. 5. Spring-loaded Tensioners - Applies constant pulling force at both ends of the cutting zone to keep the plastic strip taut and prevent wrinkling. Pill Repositioning 1. Gravity Tilt Orientation - Angles the entire feeding track (e.g., at 45 degrees) so pills naturally fall to the bottom edge of the pouches, away from the cutting line. 2. Soft Rubber Wiper - A flexible paddle that sweeps across the pouch to gently push the pill away from the perforation before the blade descends. 3. Foam Squeeze Rollers - Soft, compressible rollers that grip the pouch and naturally displace the hard pill to the side without crushing it. 4. Nylon Brush Sweeper - A row of soft bristles that brush over the strip to flatten the packaging and gently push pills aside. Cutting Mechanisms 1. Rotary Paper Trimmer - A sharp circular blade that slides laterally across a fixed track to slice the plastic cleanly. 2. Serrated Guillotine Blade - A downward-pressing blade with jagged teeth to easily pierce and shear the smooth plastic packaging. 3. Differential Speed Tearing - Uses two sets of rollers where the secondary rollers spin faster, snapping the strip along the pre-existing perforations via tension. 4. V-shaped Punch Blade - A blade shaped like a 'V' that concentrates the cutting force at the center of the strip for a cleaner, non-slip cut. Transportation / Movement 1. Dual Rubber Tank Treads - Two parallel belts that sandwich the strip from top and bottom to provide high friction and prevent slipping. 2. Edge Pinch Rollers - Small rollers that only grip the flat, heat-sealed edges of the strip, ensuring the central pill pocket is never compressed.

	3. Intermittent Geneva Drive - A mechanical gear system that translates continuous rotation into intermittent stop-and-go motion, allowing a pause for the cut. Output / Collection 4. Motorized Diverter Flap - A mechanical gate that swings left or right to sort single pouches into one bin and larger strips (e.g., 5s or 10s) into another. 5. Slanted Collection Chute - A simple gravity-fed ramp that guides the cut packages directly into standard hospital storage bins. 6. Drop-count Photoelectric Sensor - An infrared beam sensor placed at the exit that counts the falling packages to track batch sizes.
James	Cutting Mechanisms 1. Guillotine-style blades - drop from above at a specific angle 2. Rotary cutting blade - spinning wheel with blades attached at set intervals, spins to cut, adjustable speed to accommodate batch sizes 3. Scissor mechanism - two blades attached closely to act like scissors 4. Wire cutters - heated wire which burns through the plastic quickly 5. Laser cutter - small laser burns through parts of the packaging with improved precision compared to wire cutters 6. Waterjet cutter - high-pressure water to split the packages Transportation mechanisms 7. Friction rollers - rubber studs on the roller 8. Toothed rollers - teeth grab on the side of the bag 9. Vacuum conveyor - suction holds the strip to the conveyor 10. Pinching wheel - two wheels pinch the top and bottom of the packaging, “spits” the roll out 11. Lever pusher - when the batch is cut, it falls and hits a lever, which allows the next batch to advance 12. Spaced perforation rollers - rollers with set teeth at the perforations of the packages, teeth eat into the perforations and advance it forward, can also be a cutting mechanism 13. Tilted drop-chute - the machine is tilted at a specific angle, cut strips fall Pill repositioning mechanisms 14. Vibrating, tilted track - vibrates and tilts the pill near the preferred edge of the package 15. Compression rollers - gradually roll/ guide the tablet to the edge of the roller 16. Angled rails - forces the pills onto an edge 17. Camera detection - detects the position of the pill, adjusts the advance of the strip to avoid pill damage Pill output/storage

18. Stacking tray - cut strips fall into a tray with adjustable sizes to accommodate batch cut size

SCAMPER

19. Original: Banknote counting/ sorting machine
    a. Modify/ Separate: Lever arm with heated wire, cuts the pouch at given intervals
    b. Adapt/ Advance: pinching rollers on the sides
    c. Substitute/ Reposition: pills automatically drop down due to gravity
    d. Combine/ Output: output into stacks
20. Original: Label printer
    a. Modify/ Separate: modified the tear bar into a moving cutter wheel/ blade that swipes whenever a certain batch length is achieved
    b. Adapt/ Advance: Adapted the current label feeding mechanism to accommodate our dimensions
    c. Substitute/ Reposition: Combine the feeding track with an angled pill pusher/ guide
    d. Put to another use/ Output: repurpose the tray where labels fall into for an organized pouch strip stack
21. Original: Cricut machine (Vinyl cutter)
    a. Substitute/ Separate: Modify the blade to accommodate cutting plastic
    b. Adapt/ Advance: Adapt the feeding mechanism to handle long pill strips
    c. Combine/ Reposition: add a second tool which pushes the pill to the sides of the machine automatically using cameras and AI
    d. Modify/ Output: repurpose the output to have a container to hold pill strips
22. Original: Heat sealer for food
    a. Substitute/ Separate: substitute the heating bar with wire elements that can cut through plastic, include a vent to vacuum up cutting odours
    b. Adapt/ Advance: Adapted the existing conveyor system with tracks that grip the packages
    c. Combine/ Reposition: Combine compressing rollers with feed guides to reposition pills
    d. Modify/ Output: Modified the cooling section to include an output tray and to cool the cut sections, preventing melted plastic
23. Original: 3D printer
    a. Modify/ Separate: substitute the filament extruder with a low-powered laser module for cutting
    b. Substitute/ Advance: Substitute the printing bed with conveyor grips to hold the strip in place during cutting
    c. Combine/ Reposition: Repositions the laser head with the cameras
    d. Adapt/ Output: Adapt the front ledge of the printer to have an output basket

Maria	S ● Substitute fixed blades with interchangeable cutting blades. ● Substitute standard conveyor belt with roller feed system. ● Substitute manual loading with gravity-fed system. C ● Combine feeding and alignment guides into a single adjustable track. ● Combine counting mechanism + length detection. ● Combine clamping mechanism + cutter for more accurate separation. A ● Add adjustable guide rails ● Add an emergency stop and safety guard. ● Add predictive maintenance sensors. M ● Modify blade angle for cleaner cuts. ● Modify feeding angle to reduce slippage. ● Modify separation force to avoid pill damage. ● Change from cut-based separation → perforation tearing mechanism. P ● Use paper trimmer mechanism principles. E ● Eliminate multiple conveyors — use a single feed path. R ● Rearrange process to measure → clamp → cut → release → stack. ● Rotate cutting mechanism instead of moving the strip. Blue-Sky Thinking ● Vibration to push pills to the side ● Compression roller ● Angled conveyor surface ● Air pulse
Nolan	Separate tablet packages 1. Tearing 2. Hinged blade 3. Stationary blade (table saw) 4. Laser cutter Advance the tablet strip within the machine 1. Wheels 2. Conveyor belt

	Reposition tablets within the package pouch 1. Track 2. Pushing arm Output tablet strips 1. Slide

G.2: Means separated by functions

PF. 1	Separate tablet packages

1. Tearing
2. Hinged blade
3. Stationary blade (table saw)
4. Laser cutter
5. Guillotine-style blades - drop from above at a specific angle
6. Rotary cutting blade - spinning wheel with blades attached at set intervals, spins to cut, adjustable speed to accommodate batch sizes
7. Scissor mechanism - two blades attached closely to act like scissors
8. Wire cutters - heated wire which burns through the plastic quickly
9. Laser cutter - small laser burns through parts of the packaging with improved precision compared to wire cutters
10. Waterjet cutter - high-pressure water to split the packages
11. Saw
12. X-acto Knife
13. Laser Cutter
14. Punching/Stamping
15. Waterjets
16. Hot Knife
17. Freezing (causing brittleness)
18. Robot Arm
19. Offshore cutting task to Asia, China or the Philippines
20. Tearing bags
21. Machete
22. Table saw
23. Pliers
24. Heat (flame torch)
25. Interchangeable cutting blades (circular motion) instead of one blade that goes up and down
26. Adjustable blade angles
27. Tear pills apart instead of cut
28. Paper trimming mechanism

29. High speed rotary cutter 30. Abrasive discs 31. Rotary Paper Trimmer 32. Serrated Guillotine Blade 33. Uses low temperature to make the plastic bags brittle to assist separation 34. Saw-tooth blades for clean separation (either punch or slice back and forth) 35. Differential Speed Tearing 36. Ultrasonic blades (vibrating blade) 37. V-shaped Punch Blade	
SF. 1	Advance the tablet strip within the machine
1. Wheels 2. Conveyor belt 3. Robotic Arm 4. Carry by Hand 5. Use gravity to unroll (on a slant) 6. Conveyor Belt 7. Rollers 8. Robot Car 9. Vacuum 10. Floating on Water 11. Friction rollers - rubber studs on the roller 12. Toothed rollers - teeth grab on the side of the bag 13. Vacuum conveyor - suction holds the strip to the conveyor 14. Pinching wheel - two wheels pinch the top and bottom of the packaging, “spits” the roll out 15. Lever pusher - when the batch is cut, it falls and hits a lever, which allows the next batch to advance 16. Spaced perforation rollers - rollers with set teeth at the perforations of the packages, teeth eat into the perforations and advance it forward, can also be a cutting mechanism 17. Tilted drop-chute - the machine is tilted at a specific angle, cut strips fall 18. Roller feed system 19. Gravity feed system 20. Single track 21. Guide rails 22. and a roller for transporting packages 23. V-groove track	
SF. 2	Reposition tablets within the package pouch
1. Track 2. Pushing arm	

3. Vibrating, tilted track - vibrates and tilts the pill near the preferred edge of the package 4. Compression rollers - gradually roll/ guide the tablet to the edge of the roller 5. Angled rails - forces the pills onto an edge 6. Camera detection - detects the position of the pill, adjusts the advance of the strip to avoid pill damage 7. Clamps (on edges of strips) 8. Robotic Arm 9. Carry by Hand 10. Rollers (on top) 11. Robot Car 12. Vacuum 13. Floated on Water 14. Metal Plates (on edges) 15. Maintenance sensors 16. Grip on track to avoid slippage 17. Mechanism to push pills to the side 18. Vibration counterbelt to push the pill to one side 19. Angled track 20. Air pulse to encourage the pill to move to the side 21. Compression roller that will push the pill to one side	
SF. 3	Output tablet strips
1. Slide 2. Stacking tray - cut strips fall into a tray with adjustable sizes to accommodate batch cut size	

G.3: Means consolidation

Below is a list of complete and consolidated ideas, with similar or duplicate ideas removed.
Duplicates are highlighted with this colour

PF. 1	Separate tablet packages
1. Hinged blade 2. Stationary blade (table saw) 3. Laser cutter 4. Guillotine-style blades - drop from above at a specific angle 5. Rotary cutting blade - spinning wheel with blades attached at set intervals, spins to cut, adjustable speed to accommodate batch sizes 6. Scissor mechanism - two blades attached closely to act like scissors 7. Wire cutters - heated wire which burns through the plastic quickly	

8. Laser cutter - small laser burns through parts of the packaging with improved precision compared to wire cutters
9. Waterjet cutter - high-pressure water to split the packages
10. Saw
11. X-acto Knife
12. Laser Cutter
13. Punching/Stamping
14. Waterjets
15. Hot Knife
16. Freezing (causing brittleness)
17. Robot Arm
18. Tearing bags
19. Machete
20. Table saw
21. Pliers
22. Heat (flame torch)
23. Interchangeable cutting blades (circular motion) instead of one blade that goes up and down
24. Adjustable blade angles
25. Tear pills apart instead of cutting
26. Paper trimming mechanism
27. High-speed rotary cutter
28. Abrasive discs
29. Rotary Paper Trimmer
30. Serrated Guillotine Blade
31. Uses low temperature to make the plastic bags brittle to assist in separation
32. Saw-tooth blades for clean separation (either punch or slice back and forth)
33. Differential Speed Tearing
34. Ultrasonic blades
35. V-shaped Punch Blade

SF. 1	Advance the tablet strip within the machine

1. Wheels
2. Conveyor belt
3. Robotic Arm
4. Carry by Hand
5. Use gravity to unroll (on a slant)
6. Conveyor Belt
7. Rollers
8. Robot Car
9. Vacuum

10. Floating on Water 11. Friction rollers - rubber studs on the roller 12. Toothed rollers - teeth grab on the side of the bag 13. Vacuum conveyor - suction holds the strip to the conveyor 14. Pinching wheel - two wheels pinch the top and bottom of the packaging, “spits” the roll out 15. Lever pusher - when the batch is cut, it falls and hits a lever, which allows the next batch to advance 16. Spaced perforation rollers - rollers with set teeth at the perforations of the packages, teeth eat into the perforations and advance it forward, can also be a cutting mechanism 17. Tilted drop-chute - the machine is tilted at a specific angle, cut strips fall 18. Roller feed system 19. Gravity feed system 20. Single track 21. Guide rails and a roller for transporting packages 22. V-groove track	
SF. 2	Reposition tablets within the package pouch
1. Track 2. Pushing arm 3. Vibrating, tilted track - vibrates and tilts the pill near the preferred edge of the package 4. Compression rollers - gradually roll/ guide the tablet to the edge of the roller 5. Angled rails - forces the pills onto an edge 6. Camera detection - detects the position of the pill, adjusts the advance of the strip to avoid pill damage 7. Clamps (on edges of strips) 8. Robotic Arm 9. Carry by Hand 10. Rollers (on top) 11. Robot Car 12. Vacuum 13. Floated on Water 14. Metal Plates (on edges) 15. Maintenance sensors 16. Grip on track to avoid slippage 17. Vibration counterbelt to push the pill to one side 18. Angled track 19. Air pulse to encourage the pill to move to the side 20. A compression roller that will push the pill to one side	
SF. 3	Output tablet strips

1. Slide
2. Stacking tray - cut strips fall into a tray with adjustable sizes to accommodate batch cut size

G.4: Idea formulation

Linked below is the morph chart that was used to create the list of ideas found in H1 completed via Google Sheets.
https://docs.google.com/spreadsheets/d/10W6SA1ihasgjWV9eTHYyPtGM8L0IM6o1nsa9F4rWy3o/edit?usp=sharing

Appendix H: Idea Selection Process

Appendix H documents the full progression from idea generation to concept selection, moving from an initial set of 69 ideas through consolidation, feasibility evaluation, multivoting, and final design selection.
** After the selection process was completed, our three proposed alternative designs were labelled with their names to improve the readability of the idea selection process.*

H.1: Full List

This section presents the complete list of 69 ideas generated using the morph chart in Appendix G4. Duplicate concepts are highlighted in blue to make the later consolidation process more efficient and transparent. This section captures the breadth of the team's early design exploration before any filtering or evaluation.

1. **Rail-Aligned Rotary Vacuum Cutter:** A vacuum conveyor holds the strip flat while a rotary cutting blade trims through the plastic. Angled rails gradually push the pill to a safe edge, and the cut piece drops into an adjustable stacking tray.
2. **SmartLaser Pinch-Wheel Separator:** Pinch wheels move the strip forward while a camera detects pill position. A laser cuts around it, and the strip falls into a collection tray.
3. **Vision-Laser MicroCut Conveyor:** A conveyor feeds the strip under a camera-guided laser that adjusts the cut in real time. A slide directs the finished pieces into a bin.
4. **HydroJet Lever-Feed Splitter:** A lever pusher advances the strip beneath a waterjet. A small air pulse pushes the pill aside, and the cut strip falls into a tray.
5. **Ultrasonic Rail-Guided Cutter:** Ultrasonic blades cut the advancing strip on a conveyor. Angled rails push the pill away before the cut, and pieces fall into a bin.

6. **High-Frequency Ultrasonic Slide Cutter:** A conveyor brings strips beneath an ultrasonic tool that vibrates rapidly to slice the film. Plastic rails shift the pill aside, and a slide collects the pieces.
7. **Vacuum-Stabilized Rotating Blade Track:** A rotating blade slices the strip as a vacuum suction panel stabilizes movement. Two angled guide rails progressively shift pills aside before cutting and a slide collects the output.
8. **Lever-Driven Waterjet Slicer:** A high-pressure waterjet cuts strips fed by a lever mechanism, with air jets aligning the pill before slicing. Output slides down a chute.
9. **Robo-Punch Guidance System:** A robotic arm positions each pouch under a stamping punch. A vibrating tilted track ensures the pill slides toward the safe edge before cutting.
10. **Automated Punch-Robot Feeder:** A robot places the strip under a stamping die while micro-vibrations reposition pills. Segments drop into a stacking tray.
11. **Heated Wire Roller Advance Cutter (ThermoCut*):** Studded rollers push the strip beneath a heated wire that melts through the packaging. A slide guides the pieces down.
12. **Thermal Wire Melt-Line Roller Cutter:** Friction rollers feed the plastic strip into a heated cutting wire, and compression rollers push the pill toward an edge. Output pieces fall into a tray.
13. **V-Blade Vacuum Punch Station:** A vacuum belt locks the strip to the surface while a V-shaped punch slices downward. Air pulses reposition the pill.
14. **Abrasive Disc Pinch-Feed Cutter:** A rotating abrasive disc cuts the plastic while a pinch wheel feeds the strip. Compression rollers shift the pill aside.
15. **Disc-Wheel Abrasion Slicer:** A small abrasive wheel trims the strip fed by friction rollers. The pill is guided away by angled rails.
16. **Cryo-Crack Belt Separator:** A freezing nozzle chills the plastic until brittle, and a conveyor snaps it along a guide rail.
17. **Freeze-Break Rail Channel Cutter:** The strip is cooled along a rail until it fractures cleanly. A chute directs the pieces downward.
18. **Vacuum-Guided V-Punch Separator:** The strip is held by vacuum while a V-blade removes a small segment. Air jets keep the pill away from the cutting zone.
19. **Tear-Roller Differential Cutter:** Friction rollers rotate at offset speeds to tear the strip along a seam. Grip rails guide the pill aside.
20. **Rotary Trim-Rail Alignment System:** A high-speed rotary blade cuts strips while angled rails push the pill away. A slide captures output.

21. **High-Speed Rotary Rail Cutter:** A spinning cutting wheel slices the strip fed by rollers; rails shift the pill aside.
22. **Pizza-Wheel Foam Roller Cutter:** Foam rollers push the pill aside as a pizza-wheel blade slices through the packaging.
23. **Rotary Pizza-Blade Pill-Safe Trimmer:** A rolling pizza-style cutter trims plastic strips guided by foam wheels.
24. **Saw-Tooth Pinch Roller Station:** A saw-tooth blade slices strips fed by pinch rollers, with compression rollers repositioning the pill.
25. **Differential Speed Tear Station:** Different roller speeds pull the strip apart along perforations while rails align the pill.
26. **Serrated Saw-Feed Cutter:** A serrated blade cuts plastic fed by pinch rollers; rails align the pill away.
27. **Gravity Slide Punch Separator:** A slanted gravity-slide moves the strip into a punching blade. Angled rails position the pill beforehand.
28. **Vacuum-Scissor Rail Cutter:** A vacuum conveyor holds the strip while scissors cut through the packaging. Compression rollers reposition pills.
29. **Cooling Blade Gravity Dropper:** A cooling air jet chills the strip, and a vertical blade slices the weakened seam as gravity pulls it downward.
30. **Frost-Edge Gravity Cutter:** Cold air makes the plastic brittle before a static blade slices it.
31. **Dual-Roll Guillotine Feed System:** Rollers feed the strip into a guillotine blade while compression rollers shift pills away.
32. **Clamp-Stabilized Rotary Trim Unit:** Edge clamps hold the strip as a paper-trimmer style rotary cutter slices it.
33. **Trimmer-Clamp Edge-Feed System:** Clamps secure the strip as a rotary trimmer advances along the cutline.
34. **Vacuum-Corner Robotic Cutter:** A robotic arm places strips against a fixed cutting blade while a vacuum draws pills to a safe corner.
35. **Robo-Blade Vacuum Positioning Station:**A robot positions the strip and vacuum pulls pills aside before cutting.
36. **Laser-Camera Pinch Rail Station:** A laser cuts strips advanced by pinch wheels, guided by camera detection and angled rails.

37. **Slanted Punch Gravity-Feed Channel:** Strips slide down a slope into a punch blade while rails reposition the pill.
38. **Camera-Laser Rail-Guide Cutter:** A camera maps pill location for a laser cut as rails force the pill away.
39. **Roller-Advance Guillotine Assembly:** Rollers reposition pills and feed the strip directly under a guillotine.
40. **Scissor-Vacuum Alignment Cutter:** A scissor mechanism cuts the strip stabilized by vacuum suction and shifted by rails.
41. **Vacuum driven rail aligned rotary cutter:** A rotary cutting blade slices strips as they advance on a vacuum conveyor, while angled rails reposition pills against pocket edges, and finished pieces fall into an adjustable stacking tray.
42. **Guillotine Friction Compression Rollers:** A guillotine blade drops to cut strips fed by friction rollers, while compression rollers gradually guide tablets to the edge, and cut strips slide down a slide into a collection area.
43. **Laser Cutter with Pinching Conveyor:** A laser cutter burns through packaging as a pinching conveyor advances the strip, while camera detection repositions the cut to avoid pill damage, and pieces drop into a stacking tray.
44. **Lever and Air Driven Water Jet:** A waterjet cutter splits packages fed by a lever pusher mechanism, while air pulses encourage pills to move to the side, and finished strips fall onto a slide.
45. **Automated Robotic Punching Station:** A punching/stamping station cuts pouches as a robotic arm places them into position, while vibrating tilted track repositions pills, and output collects in an adjustable stacking tray.
46. **Ultrasonic Blade Fed by Gravity and Foam Roller:** Ultrasonic blades cut strips transported by gravity, while foam rollers force pills onto a safe edge, and pieces slide down a slide into a bin.
47. **Rotary Paper Trimmer with Rollers and Clamps for Positioning:** A rotary paper trimmer cuts strips fed by rollers, while clamps on edges reposition the strip for accuracy, and finished pieces fall into a stacking tray.
48. **Saw with Gravity and Grippy Track:** A saw cuts strips as they advance via gravity on a slant, while grip on track prevents slippage for repositioning, and output slides down a slide.

49. **Heated Wire with Studded Rollers:** A heated wire cutter burns through plastic as studded rollers advance the strip, while camera detection repositions the cut, and pieces collect in a stacking tray.
50. **V-Shaped Blade with Vacuum Conveyor and Air Pulse:** A V-shaped punch blade cuts strips held by a vacuum conveyor, while air pulse repositions pills, and output drops into an adjustable stacking tray.
51. **Abrasive Disc with Pinching Wheel and Compression Rollers:** An abrasive disc cuts strips fed by a pinching wheel, while compression rollers guide tablets to the edge, and pieces slide down a slide.
52. **Freeze with Conveyor Belt and Angled Rails:** A freezing mechanism embrittles and breaks plastic on a conveyor belt, while angled rails reposition the strip, and brittle pieces break into a stacking tray.
53. **Tearing with Friction Rollers:** Differential speed tearing separates strips advanced by friction rollers, while grip on track repositions the material, and torn pieces fall onto a slide.
54. **Angled Rotary Cutter:** A high-speed rotary cutter slices strips fed by rollers, while angled rails force pills to the edge, and output slides down a slide.
55. **Rotary Cutter and Wheels (PharmaSlice*):** A pizza cutter slices strips fed by roller wheels (with foam to accommodate for different pill sizes), while the wheels forces pills to the edge, and outputs into a basket
56. **Saw-Tooth Blade Cutter with Pinching Wheel Rollers:** A saw-tooth blade cuts strips fed by a pinching wheel, while compression rollers guide tablets, and finished pieces collect in a stacking tray.
57. **Punching with Gravity Feed and Angled Rails for Pill Position:** A punching/stamping station cuts strips fed by gravity on a slant, while angled rails reposition the strip, and pieces drop into a stacking tray.
58. **Scissor Vacuum with Compression Rollers:** A scissor mechanism cuts strips on a vacuum conveyor, while compression rollers guide tablets to the edge, and finished pieces drop into a stacking tray.
59. **Gravity Cooling Vertical Blade:** The strip feeds downward on a slanted path using gravity. Cooling makes the package more brittle, while metal edge plates keep it aligned. Once weakened, the package is separated via a vertical blade and the finished strip travels down the slide.

60. **Heated Wire Gravity Slide:** A hot wire cuts strips, speed fluctuating depending on the number of packages, and uses gravity to slide down the path.
61. **Rollers and Guillotine Blade:** Rollers move the strip through the system. Compression rollers push the pill away from the cutting zone. A guillotine blade cuts at a controlled angle, and the separated segment exits via a slide.
62. **Scissors with Edge Clamps and Conveyor Belt:** The conveyor carries the strip through the machine while edge clamps stabilize it. The scissor blades cut at the target location. The cut strips then move down a slide for collection.
63. **Robotic Arm:** A robotic arm that performs all means (the already existing solution but without any humans).
64. **Guillotine with Two Conveyor Belts and Foam Rollers:** Blade that cuts downwards with guardrails, made of two conveyor belts (the space in between is for the blade to pass through) and foam roller to push pills to the side
65. **Ultrasonic Blade with Friction Roller:** Friction rollers pull the strip forward while angled rails shift the pill toward a safer edge. Ultrasonic blades cut with reduced mechanical force, helping reduce deformation of the package. The finished pouch exits through a slide.
66. **Robotic Arm with Vacuum and Fixed Cutting Blade:** A robotic arm places and positions each strip against a fixed cutting blade. A vacuum mechanism moves the pills to a specific corner during the cut. The finished pieces are then deposited into a stacking tray.
67. **Pinching Wheel with Camera and Laser:** A pinching wheel moves the pouches strip forward in small increments. Camera detection identifies the pill's position and ensures the cut avoids the tablet. A laser makes a precise cut, and the finished unit drops into the tray.
68. **Rotary Paper Trimmer with Angled Rails:** Perforation rollers advance the strip by engaging repeated package features. Angled rails encourage the pill toward one side before cutting. A rotary paper-trimmer style cutter then slices the strip, and the finished piece exits on a slide.
69. **Hinged Blade with Roller (SliceFlow*):** The conveyor brings the pouch strip into the cutting zone. A roller helps reposition the pill or the strip before cutting. The hinged blade swings through the material, and the cut section exits down a slide.

H.2: Consolidation

This section contains the combined and cleaned list of ideas after removing duplicates and merging similar concepts. This consolidation step ensures that only unique, clearly differentiated ideas progress to evaluation. This resulting list forms the basis for the feasibility check that follows.

1. Rail-Aligned Rotary Vacuum Cutter
2. SmartLaser Pinch-Wheel Separator
4. HydroJet Lever-Feed Splitter
5. Ultrasonic Rail-Guided Cutter
9. Robo-Punch Guidance System
11. Heated Wire Roller Advance Cutter (ThermoCut*)
13. V-Blade Vacuum Punch Station
14. Abrasive Disc Pinch-Feed Cutter
16. Cryo-Crack Belt Separator
19. Tear-Roller Differential Cutter
20. Rotary Trim-Rail Alignment System
22. Pizza-Wheel Foam Roller Cutter
24. Saw-Tooth Pinch Roller Station
27. Gravity Slide Punch Separator
28. Vacuum-Scissor Rail Cutter
29. Cooling Blade Gravity Dropper
31. Dual-Roll Guillotine Feed System
32. Clamp-Stabalized Rotary Trim Unit
34. Vacuum-Corner Robotic Cutter
36. Laser-Camera Pinch Rail Station
41. Vacuum driven rail aligned rotary cutter
44. Lever and Air Driven Water Jet:
45. Automated Robotic Punching Station
46. Ultrasonic Blade Fed by Gravity and Foam Roller
47. Rotary Paper Trimmer with Rollers and Clamps for Positioning
48. Saw with Gravity and Grippy Track
49. Heated Wire with Studded Rollers
50. V-Shaped Blade with Vacuum Conveyor and Air Pulse
51. Abrasive Disc with Pinching Wheel and Compression Rollers
52. Freeze with Conveyor Belt and Angled Rails
53. Tearing with Friction Rollers
54. Angled Rotary Cutter
55. Rotary Cutter and Wheels (PharmaSlice*)
56. Saw-Tooth Blade Cutter with Pinching Wheel Rollers
57. Punching with Gravity Feed and Angled Rails for Pill Position
58. Scissor Vacuum with Compression Rollers
59. Gravity Cooling Vertical Blade
60. Heated Wire Gravity Slide

62. Scissors with Edge Clamps and Conveyor Belt
63. Robotic Arm
66. Robotic Arm with Vacuum and Fixed Cutting Blade
69. Hinged Blade with Roller (SliceFlow*)

H.3: Feasibility Check

This section evaluates each consolidated idea against the system's functions and constraints. Ideas that meet all requirements are highlighted in green, while those that fail one or more criteria appear in red and are removed from further consideration. This step ensures that only viable concepts continue to the multivoting stage.

Table H.3.1 The Feasibility Check Process

Idea	PF.1	SF.1	SF. 2	SF. 3	C. 1	C. 2	C. 3	C. 4	C. 5
1. Rail-Aligned Rotary Vacuum Cutter	Y	Y	Y	Y	Y	Y	Y	Y	Y
2. SmartLaser Pinch-Wheel Separator	Y	Y	Y	Y	Y	Y	Y	Y	N
4. HydroJet Lever-Feed Splitter	Y	Y	Y	Y	Y	N	Y	Y	Y
5. Ultrasonic Rail-Guided Cutter	Y	Y	Y	Y	Y	Y	Y	Y	Y
9. Robo-Punch Guidance System	Y	Y	Y	Y	Y	Y	Y	Y	N
11. Heated Wire Roller Advance Cutter (ThermoCut*)	Y	Y	Y	Y	Y	Y	Y	Y	Y
13. V-Blade Vacuum Punch Station	Y	Y	Y	Y	Y	Y	Y	Y	Y
14. Abrasive Disc Pinch-Feed Cutter	Y	Y	Y	Y	Y	Y	Y	Y	Y
16. Cryo-Crack Belt Separator	Y	Y	Y	Y	Y	N	Y	Y	N
19. Tear-Roller Differential Cutter	Y	Y	Y	Y	Y	Y	Y	Y	Y
20. Rotary Trim-Rail Alignment System	Y	Y	Y	Y	Y	Y	Y	Y	Y
22. Pizza-Wheel Foam Roller Cutter	Y	Y	Y	Y	Y	Y	Y	Y	Y
24. Saw-Tooth Pinch Roller Station	Y	Y	Y	Y	Y	Y	Y	Y	Y
27. Gravity Slide Punch Separator	Y	Y	Y	Y	Y	Y	Y	Y	Y
28. Vacuum-Scissor Rail Cutter	Y	Y	Y	Y	Y	Y	Y	Y	Y
29. Cooling Blade Gravity Dropper	Y	Y	Y	Y	Y	N	N	Y	N

31. Dual-Roll Guillotine Feed System	Y	Y	Y	Y	Y	Y	Y	Y	Y
32. Clamp-Stabalized Rotary Trim Unit	Y	Y	Y	Y	Y	Y	Y	Y	Y
34. Vacuum-Corner Robotic Cutter	Y	Y	Y	Y	Y	Y	Y	Y	N
36. Laser-Camera Pinch Rail Station	Y	Y	Y	Y	Y	Y	Y	Y	N
41. Vacuum driven rail aligned rotary cutter	Y	Y	Y	Y	Y	Y	Y	Y	N
44. Lever and Air Driven Water Jet:	Y	Y	Y	Y	Y	N	Y	Y	N
45. Automated Robotic Punching Station	Y	Y	Y	Y	Y	Y	Y	Y	N
46. Ultrasonic Blade Fed by Gravity and Foam Roller	Y	Y	Y	Y	Y	Y	Y	Y	Y
47. Rotary Paper Trimmer with Rollers and Clamps for Positioning	Y	Y	Y	Y	Y	Y	Y	Y	Y
48. Saw with Gravity and Grippy Track	Y	Y	Y	Y	Y	Y	N	Y	Y
49. Heated Wire with Studded Rollers	Y	Y	Y	Y	Y	Y	Y	Y	Y
50. V-Shaped Blade with Vacuum Conveyor and Air Pulse	Y	Y	Y	Y	Y	Y	Y	N	Y
51. Abrasive Disc with Pinching Wheel and Compression Rollers	Y	Y	Y	Y	Y	N	Y	Y	Y
52. Freeze with Conveyor Belt and Angled Rails	Y	Y	Y	Y	Y	Y	Y	Y	Y
53. Tearing with Friction Rollers	Y	Y	Y	Y	Y	N	Y	Y	N
54. Angled Rotary Cutter	Y	Y	Y	Y	Y	Y	Y	Y	Y
55. Rotary Cutter and Wheels (PharmaSlice*)	Y	Y	Y	Y	Y	Y	Y	Y	Y
56. Saw-Tooth Blade Cutter with Pinching Wheel Rollers	Y	Y	Y	Y	Y	N	N	Y	Y
57. Punching with Gravity Feed and Angled Rails for Pill Position	Y	Y	Y	Y	Y	N	Y	Y	Y
58. Scissor Vacuum with Compression Rollers	Y	Y	Y	Y	Y	Y	Y	Y	N
59. Gravity Cooling Vertical Blade	Y	Y	Y	Y	Y	Y	N	Y	N
60. Heated Wire Gravity Slide	Y	Y	Y	Y	Y	Y	Y	Y	Y
62. Scissors with Edge Clamps and Conveyor	Y	Y	Y	Y	Y	Y	Y	Y	Y

Belt									
63. Robotic Arm	Y	Y	Y	Y	Y	Y	Y	Y	N
66. Robotic Arm with Vacuum and Fixed Cutting Blade	Y	Y	Y	Y	Y	Y	Y	Y	N
69. Hinged Blade with Roller (SliceFlow*)	Y	Y	Y	Y	Y	Y	Y	Y	Y

H.4: Multivoting

This section shows the multivoting procedure used to narrow the feasible ideas down to a

Table H.4.1. The Multivoting Process

Round 1		**Round 2**		**Final 9 Ideas**
Idea	**Voting**	**Idea**	**Voting**	
1. Rail-Aligned Rotary Vacuum Cutter	II	1. Rail-Aligned Rotary Vacuum Cutter	I	
5. Ultrasonic Rail-Guided Cutter	II	5. Ultrasonic Rail-Guided Cutter	II	5. Ultrasonic Rail-Guided Cutter
11. Heated Wire Roller Advance Cutter (ThermoCut*)	I	11. Heated Wire Roller Advance Cutter (ThermoCut*)	III	11. Heated Wire Roller Advance Cutter (ThermoCut*)
13. V-Blade Vacuum Punch Station				
14. Abrasive Disc Pinch-Feed Cutter	II	14. Abrasive Disc Pinch-Feed Cutter	I	
19. Tear-Roller Differential Cutter				
20. Rotary Trim-Rail Alignment System				
22. Pizza-Wheel Foam Roller Cutter	II	22. Pizza-Wheel Foam Roller Cutter	II	22. Pizza-Wheel Foam Roller Cutter
24. Saw-Tooth Pinch Roller Station				
27. Gravity Slide Punch Separator	I	27. Gravity Slide Punch Separator	I	
28. Vacuum-Scissor Rail Cutter				
31. Dual-Roll Guillotine Feed System	II	31. Dual-Roll Guillotine Feed System	II	31. Dual-Roll Guillotine Feed System

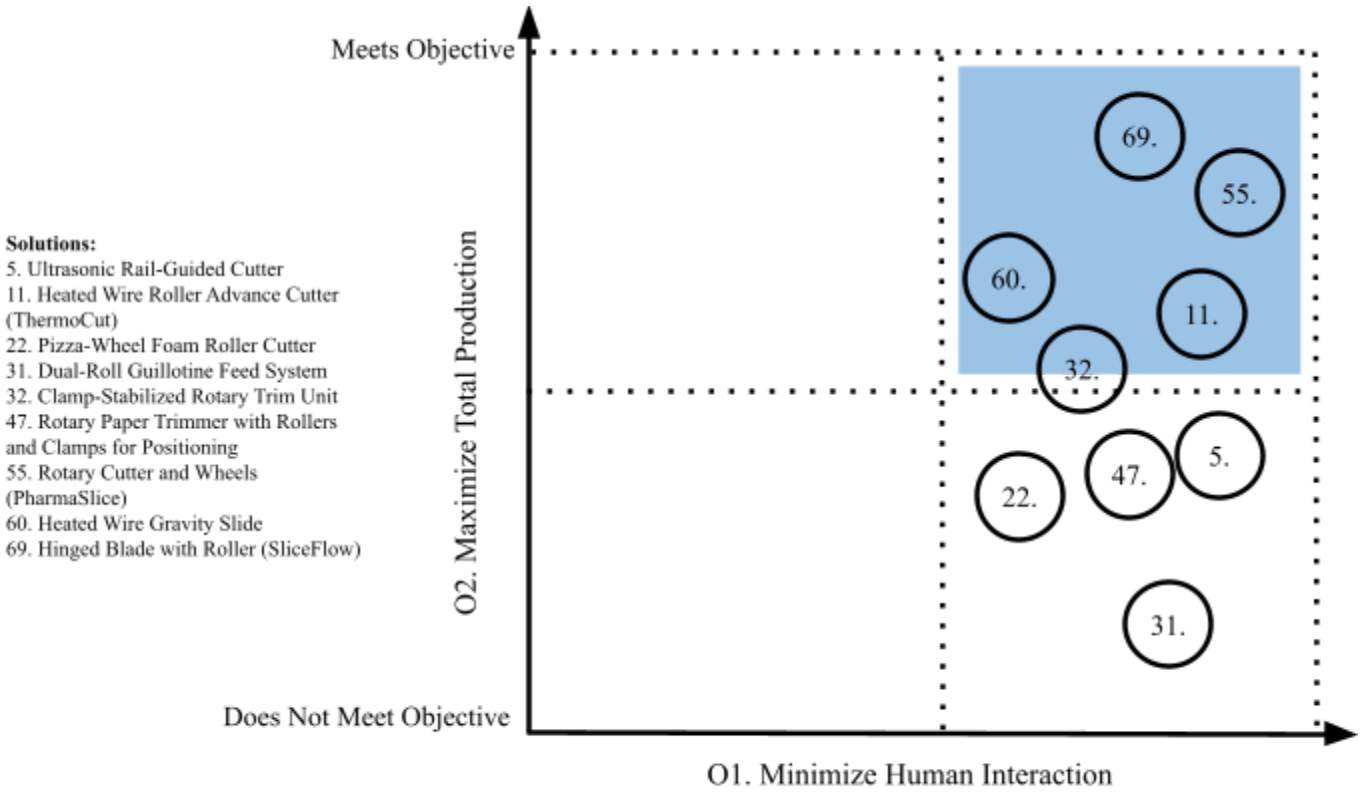
32. Clamp-Stabalized Rotary Trim Unit	II	32. Clamp-Stabalized Rotary Trim Unit	II	32. Clamp-Stabalized Rotary Trim Unit
46. Ultrasonic Blade Fed by Gravity and Foam Roller	II	46. Ultrasonic Blade Fed by Gravity and Foam Roller	I	
47. Rotary Paper Trimmer with Rollers and Clamps for Positioning	II	47. Rotary Paper Trimmer with Rollers and Clamps for Positioning	II	47. Rotary Paper Trimmer with Rollers and Clamps for Positioning
49. Heated Wire with Studded Rollers	II	49. Heated Wire with Studded Rollers	I	
52. Freeze with Conveyor Belt and Angled Rails				
55. Rotary Cutter and Wheels (PharmaSlice*)	III	55. Rotary Cutter and Wheels (PharmaSlice*)	IIIIII	55. Rotary Cutter and Wheels (PharmaSlice*)
60. Heated Wire Gravity Slide	II	60. Heated Wire Gravity Slide	III	60. Heated Wire Gravity Slide
62. Scissors with Edge Clamps and Conveyor Belt	I	62. Scissors with Edge Clamps and Conveyor Belt		
69. Hinged Blade with Roller (SliceFlow*)	IIII	69. Hinged Blade with Roller (SliceFlow*)	III	69. Hinged Blade with Roller (SliceFlow*)

Final list after multivoting:

5. Ultrasonic Rail-Guided Cutter
11. Heated Wire Roller Advance Cutter (ThermoCut*)
22. Pizza-Wheel Foam Roller Cutter
31. Dual-Roll Guillotine Feed System
32. Clamp-Stabalized Rotary Trim Unit
47. Rotary Paper Trimmer with Rollers and Clamps for Positioning
55. Rotary Cutter and Wheels (PharmaSlice*)
60. Heated Wire Gravity Slide
69. Hinged Blade with Roller (SliceFlow*)

H.5: Graphical Design Selection

Section H5 presents the graphical comparison of the nine remaining ideas against the project's top two objectives. The three highest-scoring concepts then advanced to the Pugh Chart for the final evaluation step.



Solutions:
5. Ultrasonic Rail-Guided Cutter
11. Heated Wire Roller Advance Cutter (ThermoCut)
22. Pizza-Wheel Foam Roller Cutter
31. Dual-Roll Guillotine Feed System
32. Clamp-Stabilized Rotary Trim Unit
47. Rotary Paper Trimmer with Rollers and Clamps for Positioning
55. Rotary Cutter and Wheels (PharmaSlice)
60. Heated Wire Gravity Slide
69. Hinged Blade with Roller (SliceFlow)

Appendix I: Pugh Method

Below is the pugh method used to compare each design against a datum (Design 1/PharmaSlice) across the seven project objectives, using a scoring scale from -2 to +2. Each design's relative performance was evaluated objectively, and the design with the highest overall score was selected as the final proposed solution.

Table I.1 The Pugh Chart

Objective	**Design 1 (PharmaSlice /Datum)**	**Design 2 (ThermoCut)**	**Design 3 (SliceFlow)**
O.1 Human Interaction	S	-2	0
O.2 Total Production	S	0	0
O.3 Success Rate	S	0	0
O.4 Loading Capacity	S	0	0
O.5 Maintenance Frequency	S	-1	-1
O.6 Internal Visibility from Outside	S	0	-1
O.7 Dimensions	S	+1	0
O.8 Spending	S	-1	0
Total	**0**	**-3**	**-2**

Justification:

O.1 Human Interaction:

- ThermoCut: As you have to set up the strip in between the two wheels, it will take longer to set up than the datum where only one pouch needs to be fed.

- SliceFlow: Follows the same loading mechanism as the datum therefore it has score 0.

O.2 Total Production

- This metric is unable to be objectively compared as it would need to be physically tested for both designs.

O.3 Success Rate

- This metric is unable to be objectively compared as it would need to be physically

tested for both designs.

O.4 Loading Capacity

- The loading capacity of all designs are limited by the length of the strip and therefore all the same.

O.5 Maintenance Frequency

- ThermoCut: As the design relies on heat to cut through the material, the blade will not dull over time and therefore will not need to be replaced as often. However filter replacement is crucial for health concerns.

- Slice Flow: As the design uses a flat blade it has less surface area and therefore degrades faster than a round blade.

O.6 Internal Visibility from Outside

- ThermoCut : As the cutting system is within a shell (to provide safety), there is limited visibility.

- Slice Flow: This design uses a single flat blade rather than a full 360 degree blade meaning that there is less cutting surface area and would therefore dull faster / need to be replaced more often.

O.7 Dimensions

- ThermoCut: H: 60 > 30.5cm , W: 25 < 30.5cm, L: 25<30.5cm therefore it is smaller than the datum.

- SliceFlow: H: > 30.5cm , W: 24 < 30.5cm, L: 40 > 30.5cm therefore the overall size is similar to the datum.

O.8 Spending

- ThermoCut: As this design requires a power supply to heat the wire, it will cost more than the datum.

- Slice Flow: Both designs will cost approximately the same amount as the main materials used are the same but the configuration differs.

Appendix J.1:

Justification for each objective analysis for design 1.

Objective	Analysis
Human Interaction	The design only requires the pharmacy technicians to load the strip into the machine's intake and collect the separated packages from the output basket, with occasional clearance of jams if they occur. The wheel advances packages automatically through the cutting zone, while the Arduino

	motor triggers the cutter at precise intervals to cut exactly 2, 5 or 10 packages. Each strip contains anywhere from 100 to 500 pouches, at 8000 packages a day, which will require 16-80 actions a day. Taking a rough estimate of 30 seconds per load and calibration, the max loading time would be 40 minutes. Allowing 15 minutes for occasional jams, this design stays under the 60 minutes required of O.1.
Total Production	Production rate is determined by the roller advancement speed and the rotary cutter wheel speed. The roller system advances strips at a maximum of 12.57 cm/s (based on a standard 60 RPM motor with a 40mm drive pulley), advancing one pouch length (6 cm) in roughly 0.5 seconds. The rotary cutter wheel is also powered by the same motor, with a wheel diameter of 45mm, the circumference is 14.14cm. To cut across the 4 cm package width, the wheel requires approximately 0.28 seconds to complete one cutting pass. Total cut cycle time is approximately 0.28 seconds, with no pre-heating or cooling required. Detailed production specifications for each batch size are below: ● **2-package batches**: with a strip length of 12 cm, feed time is 1.0 second plus 0.28 second cut cycle, totaling 1.28 seconds per cycle. This yields 2813 strips per hour (5,625 pouches/hour), exceeding the 1750 strips/hour requirement and completing the 8000-pouch daily goal in 1.4 hours. ● **5-package batches**: with a strip length of 30 cm, feed time is 2.5 seconds plus 0.28 second cut cycle, totaling 2.78 seconds per cycle. This yields 1295 strips per hour (6475 pouches/hour), below the 1750 strips/hour requirement, completing the 8000-pouch daily goal in 1.2 hours. ● **10-package batches**: with a strip length of 60 cm, feed time is 5.0 seconds plus 0.28 second cut cycle, totaling 5.28 seconds per cycle. This yields 682 strips per hour (6818 pouches/hour), below the 1750 strips/hour requirement, completing the 8000-pouch daily goal in 1.2 hours.
Success Rate	Guide rails constrain the strip to a precise path, keeping the pouches aligned horizontally. Friction from the advancement wheels provides consistent advancement without slippage, ensuring each cut occurs at the perforation consistently. The Arduino can be calibrated to optimize cutting timing for different cutting intervals, allowing fine adjustments to reduce the risk of misaligned cuts. With these physical and

	technical constraints and the ability to calibrate, the design is projected to achieve at least a 99.9% success rate, exceeding the objective requirement.
Loading Capacity	The open feed path has no physical stop or length limitation; strips can be fed until the entire roll is processed. The drive wheels maintain consistent grip regardless of strip length, while the guide rails and friction rollers prevent bunching, even on long strips. This allows a full input roll of 500 packets, meeting objective requirements.
Maintenance Frequency	All components are available off the shelf and are easily replaceable without specialized tools. The rotary trimmer blade is the component with the most potential wear. Industrial-grade slitter blades used for lightweight cloth fibreglass last up to 20000m worth of cutting length, at an average thickness of 100 microns, similar to our material thickness. At 4000 cuts a day for 2-package batches, this extends the blade life to over 125 days (4 months). [36][37]. Brushless DC motors are rated for up to 20000 hours of operation, taking the average at 10000 hours, working for a maximum of 3 hours a day; this is over 9 years of operation. [38].
Internal Visibility from Outside	The open-frame construction leaves the entire cutting mechanism and strip path exposed to view; however, it could be optionally covered up during operation. The design would have a box around the cutting blade, protecting the operator from the exposed blade. This visibility allows for early detection of jams or clearance of jams.
Dimensions	The prototype would have a 305mm x 305mm footprint with its base. Well within the dimension objectives, allowing placement on any standard pharmacy countertop 630mm - 762mm [39]. This compact size also allows for easy relocation if the pharmacy layout changes.
Spending	The total cost of materials will be under $200 for the prototype, with the final design benchmarked using a similar vinyl cutter machine priced at $650 at the time of writing (March 20, 2026) [40]. This leaves over $9000 dollars to modify the machine or to specifically tailor it to our needs.

Appendix J.2:

Justification for each objective analysis for design 2.

Objective	Analysis
Human Interaction	The ThermoCut design requires the operator to load strips into the machine and collect periodically from the collection basket. The DC motor-driven wheels automates strip advancement through the cutting zone, with a dual wheel design to ensure sufficient control over the packets, preventing slipping. The heated wire cycles at speeds corresponding to intervals of 2, 5, or 10 package lengths, determined from prototyping data. To avoid complexities arising from more wheels used in aligning the strips, each batch must be manually loaded and slotted between the wheels, as the heated wire with limited roller mechanism does not accommodate continuous roll feeding as seamlessly as the other two designs. Based on 8000 packages per day, at an average 100-500 packages a strip, the operator would load 16-80 packages per day. At 45 seconds per load, due to manual alignment, total loading time is 60 minutes. Adding in 20 minutes to collection and resolving occasional jams, this brings a max daily total to 80 minutes, exceeding the 60 minute objective target.
Total Production	Production rate is limited by the roller advancement speed and wire cycle times. The roller system advances strips at a maximum of 12.57 cm/s (based on a standard 60 RPM motor with a 40mm drive pulley), advances one pouch length (6 cm in length) in roughly 0.5 seconds. The heated wire cuts at a rate of 400mm/min, 6.7mm per second [25]. At a thickness of 100 microns, it takes 0.015 seconds to cut, and 1 second to cool down and heat up, for a total cycle time of approximately 1.015 seconds. Detailed production specifications for each batch size are listed below: ● **2-package batches**: With a strip length of 12 cm, feed time is 1.0 second plus 1.015 second cut cycle, totaling 2.015 seconds per cycle. This yields 1788 strips per hour (3576 pouches/hour), exceeding the 1750 strips/hour requirement and completing the 8000-pouch daily goal in 2.2 hours. ● **5-package batches**: With a strip length of 30 cm, feed time is 2.5 seconds plus 1.015 second cut cycle, totaling 3.515 seconds per cycle. This yields 1024 strips per hour (5120 pouches/hour), below the 1750 strips/hour requirement, completing the 8000-pouch daily goal in 1.6 hours. ● **10-package batches**: With a strip length of 60 cm,

	feed time is 5.0 seconds plus 1.015 second cut cycle, totaling 6.015 seconds per cycle. This yields 598 strips per hour (5980 pouches/hour), below the 1750 strips/hour requirement, completing the 8000-pouch daily goal in 1.3 hours.
Success Rate	Success rate is determined by the guide rails and precise temperature control. Internal guide rails constrains the strips to a path, maintaining alignment within ± 1mm of the cutting zone throughout advancement. The nichrome wire is heated to 300 degrees, 70 degrees above the melting point of PET, ensuring clean cuts through packaging without significant contact force [26]. This temperature margin accounts for heat dissipating to the surrounding material to ensure consistent cuts even with variations in plastic thickness. The heated wire melts through the perforated seam instead of cutting or tearing, reducing the risk of package damage. With these design features, the expected success rate is to be above 99.9%.
Loading Capacity	The roller intake has no physical stop or length restriction. Strips can be loaded end to end, though manual alignment is required for each new strip. The ATP machine's maximum output of 500 packets per roll can be loaded and processed continuously, meeting objective requirements.
Maintenance Frequency	The nichrome wire is the primary wear component in this design. Nichrome has a maximum operating temperature of 1050 degrees [27], well above the 300 degree service temperature, ensuring the wire remains within its limits and does not degrade rapidly. Under continuous operation at 300 degrees, significantly lower than the rated temperature, nichrome wire only requires replacement when it becomes oxidized, upwards of thousands of hours of operation [41]. The filter should be replaced every 3 to 12 months, as PET heated to 200-300°C is expected to release only acetaldehyde and formaldehyde while retaining high thermal stability till 382°C (< 0.4 wt.% loss) [42] [43], any hydrophilic zeolite filters selected for these aldehydes should be replaced conservatively based on prototype testing [44]
Internal Visibility from Outside	A 130mm x 180mm transparent glass panel is mounted on the side of the enclosure, providing a viewing window into the cutting zone. This window allows operators to see the strip alignment and the wire's operations. Operators can see

	only the critical cutting area but not the entire strip path due to the complex internals. The glass panel is heated and coated to prevent fogging to vaporized plastics.
Dimensions	The ThermoCut design integrates all components into a compact footprint, within 305mm x 250mm, and a height of 250mm. This is well within the dimension objectives, allowing placement on any standard pharmacy countertop 630mm - 762mm [45]. This compact size also allows for easy relocation if the pharmacy layout changes.
Spending	Nichrome wires, hydrophilic zeolite filters and small motors are easily accessible at low cost on online platforms. [46] Using aluminum as a benchmark casing material (except ceramic casing around wire for thermal protection), the prototype is expected to cost less than $600 CAD as of March 20, 2026 with materials, components, and processing accounted for. [47]

Appendix J.3:

Justification for each objective analysis for design 3.

Objective	Analysis
Human Interaction	The SliceFlow design requires the pharmacy technicians to only periodically load the strip and collect finished strips. The dual-conveyor system automates advancement through the cutting zone, while the motor-controlled blades cycle automatically at intervals corresponding to 2, 5, or 10 package lengths. The cleated slider and foam roller ensure consistent package and pill positioning without manual adjustment. Based on 8000 packages a day, and an average of 100 to 500 packages a strip, the operator loads 16 to 80 times a day. At an upper limit of 30 seconds per load, total loading time maxes out at 40 minutes. The foam roller and conveyor system operates continuously without intervention, and output packages collect in a bin, requiring collection only after needed or filled. Clearing occasional jams, estimated to be 10 minutes daily, brings the total interaction time to 50 minutes, meeting the objective goal of $\leq$ 60 minutes.

Total Production	Production rate is determined by conveyor advancement speed and blade cycle time. The conveyor belt system advances strips at a max of 12.57 cm/s (based on a standard 60 RPM motor with a 40mm drive pulley), positioning each cutting strip under 1 second each. The motor-controlled blade descends for 0.2 seconds, cuts for 0.1, and retracts for 0.2, for a total cut cycle of 0.5 seconds. The dual-conveyor design allows the next package to advance while the blade retracts, allowing for the next package to advance while the blade retracts, overlapping interactions. Each cycle, with the full advance and cutting sequence, timing expectations are below, using 6 cm for each pouch length: ● **2-package batches**: With a strip length of 12 cm, the feed takes 1 s plus a 0.5 s cut cycle; set at 1.5 seconds per cycle, this yields 2400 strips/hour (4800 pouches/hour), exceeding requirements and completing the 8000-pouch goal in 1.7 hours. ● **5-package batches**: With a strip length of 30 cm, the feed takes 2.5 s plus a 0.5 s cut cycle; set at 3 seconds per cycle, this yields 1200 strips/hour (6000 pouches/hour), below requirements, but still completing the 8000-pouch daily goal faster than the 2-package batches, in 1.3 hours. ● **10-package batches**: With a strip length of 60 cm, the feed takes 5 s plus a 0.5 s cut cycle; set at 5.5 seconds per cycle, this yields 654 strips/hour (6545 pouches/hour), below requirements, but still completing the 8000-pouch daily goal faster than the 2-package batches, in 1.2 hours.
Success Rate	Success rate is controlled by the pill repositioning system and package alignment. Before reaching the cutting station, the foam roller rotates to gently sweep each pill towards the side of its package. The roller's foam compresses slightly (3-5mm) to accommodate pill thickness while applying just enough force to reposition without crushing. This ensures pills are clear of the blade path. The cleated slider path uses low ridges that guide the edges of the pouches, maintaining alignment within ±0.5mm from left to right shifting. The conveyor gap (20 mm) allows the blade to pass completely through the package without contacting the conveyor surface, ensuring clean cuts. The acrylic cover allows for visibility while protecting the operator against cutting fragments. With these design features, the theoretical maximum failure rate from misalignment is low, with misaligned blades even lower, suggesting a high success rate

	exceeding that of the objective.
Loading Capacity	The conveyor belt intake is open with no physical stop or length restriction. Strips can be loaded end-to-end or fed from a roll. The conveyor belts (minimum width 42mm) and cleated slider bed maintain consistent traction regardless of slip length, preventing bunching or slipping during long runs. A full 500 pouch roll can be loaded at once and processed without operator intervention, meeting objective requirements.
Maintenance Frequency	As with design 1, SliceFlow will use a DC motor to power the cutting mechanism. These motors are rated for up to 20000 hours (9 years) of operation before needing replacement [48]. The limiting factor will be the dulling of the blade with wear. The average lifespan of blades used for cutting paper and plastic films is 80,000–100,000 cuts before noticeable wear. With a maximum 8,000 pouches (or 4,000 cuts) per day, each blade would last around 3 weeks before any noticeable wear [49]. However, it would take nearly 6-12 months before there are any critical drops in performance [50].
Internal Visibility from Outside	Plexiglass was chosen for its high optical clarity, strength and stiffness [51] in order to ensure staff safety while preserving full visibility of internal mechanical components and operations. The removable acrylic cover provides 360-degree visibility of the entire operation. Unlike designs with limited viewing windows, the SliceFlow's transparent enclosure allows the operator to see all internal mechanical parts. This exceeds the 180 mm visibility requirement substantially. The clear acrylic material retains over 90% of the light passed through it, allowing for little distortion when viewing. This allows for early detection of jams or misaligned packages, leading to quick intervention [52].
Dimensions	The SliceFlow design footprint is determined by the conveyor belt system; total length is 40 cm, including the input area, the cutting station, and the output chute. Total width is expected to be around 24cm, allowing for mechanical parts to fit within the frame. The complete unit dimensions are 40 cm L x 24 cm W x 15 cm H, well within the objective limits of 80 cm by 62 cm. This compact

	footprint allows placement on any standard pharmacy countertop [45].
Spending	The total cost of materials will be well under $10000 CAD. Benchmarking off of a tabletop wood saw, which has a more expensive cutting mechanism, but lacks the conveyor mechanism, costs around $500 CAD as of March 20, 2026 [53].

Appendix K:

Additional information and justifications for the methods of success.

Appendix K.1: Selected Objective and Justification

The team's top critical objectives for validating design success, ranked through pairwise comparison (Appendix E) and client feedback (Sinai Health Pharmacy, private communication, Mar. 6, 2026), are detailed below:

- O.1: Minimize Human Interaction With Design (< 60 min/day)
- O.2: Maximize total production (≥1750 strips/hour)
- O.3: Maximize success rate (≥ 99.9% undamaged packages)

Three objectives are functionally dependent on each other in the proposed design and in alignment with the client's requests. Testing them separately would fail to capture the design's real-world performance requirements.

These interdependent design objectives tie back to each function. The design must advance the strip precisely (connects to O.2), secure packages without crushing pills (O.3) and cleanly cut at the perforation line (O.2, O.3). Failures are directly linked to all three objectives:

- **Opened packages due to misaligned cuts** (O.3 failure) require human intervention to remove the damaged medication and reset the machine, increasing human interaction time (O.1) and reducing production (O.2)
- **Misalignment or feeding errors** (O.3 failure) create inconsistent cut quality, damaging the quality of medication, resulting in waste and additional human interactions (O.1).
- **Mechanical jams from slippage or bunching** (O.2 failure) stop production completely, forcing technicians to manually clear the machine, increasing human interactions (O.1) and potentially damaging packages (O.3)

The client's main problem with repetitive strain injuries is addressed by O.1; however, O.2 and O.3 are prerequisites. A design that doesn't meet the production quotas or damage packages will

not be used; therefore, the design must compare to all three objectives to ensure it is practical for pharmacy use.

Appendix K.2: Test Method Overview

The test will employ bench-top testing using the team's functional prototype of the rotary paper trimmer design. This method was selected as it allows for direct measurement of all three interdependent objectives under simulated working conditions, using hardware that would be later used in the pharmacy. [https://prolabfitouts.com.au/blog/what-is-benchtop-testing/]

Bench-top testing is a standard validation approach used in medical devices and biomedical engineering, where products are tested on a small scale in a controlled setting to verify the key mechanisms and functional performance before being deployed. This method is essential in ensuring the design meets our testing objectives to identify key issues early on. For our prototype, bench-top testing is appropriate, as it allows for direct assessment of cutting precision, speed, and required maintenance. [32][33]

Prototype Configuration:

- Rotary paper trimmer mounted on a wooden rail
- Wheel-driven advancement of the strip along a guided path for precise positioning
- Computer-controlled cutting intervals during testing (final design will use buttons or dial)
- Computer control ensures consistency between tests, allowing the team to focus on evaluating performance objectives

Test Personnel:

- Two team members will conduct testing:
    a. Member 1: Manages machine operation (loading strips, clearing jams)
    b. Member 2: Tracks time and records data
- A two-person team allows for precise documentation while maintaining consistent and safe operations [34]

Test Materials:

- Authentic hospital medical strips (~200 packages) provided by the client
- Office letter paper, roughly 110-112µm [54]
    - LDPE film: 38 ± 2µm
    - PET layer: 12 ± 2µm
    - Total one-sided thickness: 110-118µm (verified per ASTM D6988)(Sinai Health Pharmacy, private communication, Feb. 2, 2026)

According to the plastic specifications provided by the client, the pouches are made from LDPE and PET with a combined thickness of 110-118 µm (both layers) (Sinai Health Pharmacy, private communication, Feb. 2, 2026). According to Health Canada (GUI-0029), medical device

validation shall be conducted using the actual material the device will encounter or other highly similar samples when authentic samples are limited. [35].

Appendix K.3: Test Equipment, Procedure and Data Collection

Prototype Components:

- Wooden frame prototype
- 2 DC gear motors (12V, 60RPM)
    - Motor 1: drives the paper trimmer cart along the rail
    - Motor 2: Drives the wheel to advance strips
- L298N 2A motor driver
- Arduino UNO R3
- Breadboard
- Jumper wires (M-M, M-F)
- 22 AWG solid core wire
- Plywood sheets (¼” thick, 12”x12”)
- EVA black foam (½” thick, 12”x12”)

Testing Materials:

- Authentic hospital medication strips (~200 provided)
- Standard office paper (~40 sheets)

Measurement and Data Collection:

- A smartphone for video and a stopwatch
- Digital table for data collection

Output:

- Collection basket for separated packages

The prototype will be constructed on-site, using pre-purchased materials and on-site tools and machinery.

Test preparation:

1. Cut paper into 5 equal-width strips, connect strips lengthwise with tape, and prepare roughly 50 pouch lengths per strip, 10 strips total.
2. Use leftover material to run calibration tests:
    a. Verify motor 2 advances the correct distance, no slipping
    b. Verify motor 1 drives the cart trimmer smoothly

The calibration process ensures consistent and repeatable cutting distances before data collection begins, allowing the team to pinpoint key issues if they arise.

Test Execution (9 runs total, 3 for each cutting interval, twice with paper, once with strips)

1. Load the strip into the machine, ensure drive wheel is properly engaged with strip
2. Start the stopwatch and machine together
    a. Record timestamps of irregularities
3. If jam occurs, stop the machine, identify flaw, fix flaw and resume testing
4. At end of run, stop stopwatch, and calculate the number of packages processed per unit time
5. Check for any misaligned cuts, torn packages, or calibration issues in the long run
6. Repeat for 9 runs

Table K.1 details the data collected during each run and the measurement method

Table K.1: Data to be collected per run

Objective	Data collected	Measurement Method
O.1: Human Interaction	Loading time per strip	Video timestamp
	Intervention time per jam	Video timestamp
	Total run interaction time	Sum of all action times
O.2: Production Rate	Total run time per strip	Video timestamp
	$Package\ per\ minute\ = \frac{Total\ Packages}{Total\ runtime}$	Calculated
	$Proj.\ hourly\ rate\ = Pkg\ per\ min\ *\ 60$	Calculated
O.3: Success Rate	Total packages processed	Count
	Number of failures	Visual inspection + video
	$Success\ rate\ = 100\%\ *\ \lvert\frac{total-failure}{total}\rvert$	Calculated post-run

Data collection table example: screenshots are taken; however, the file is linked here
TUT#0135_Team181_CDS_Data_Collection_Template.xlsx

Master Data Log:

Run #	Date	Operators	Strip Material	Start Time	End Time	Total Pkgs	Total Run Time (min)	Total Int Time (sec)	FC-1	FC-2	FC-3	FC-4	FC-5	FC-6	FC-7	FC-8	Total Failures	Notes	Success Rate (%)	Pkgs per Minute	Hourly Rate	Daily Interaction (min)
EXAMPLE	2026-03-23	James/ Alba	Paper	10:00	10:04:30	120	4.5	45	0	0	0	0	1	0	0	0	1	Minor jam at roller, cleared quickly	99.17	26.7	1600	50.0

Event Log:

Run #	Timestamp (mm:ss)	Failure Code	Description	Duration (sec)
EXAMPLE	1:23	FC-5	Strip jammed at entry roller	15
EXAMPLE	3:47	FC-3	Cut 3mm off perforation line	0
EXAMPLE	5:12	FC-1	Pill crushed in package #42	0

Appendix K.4: Test schedule

The test will be conducted over three weeks following the CDS submission. Table K.2 details the tasks required for each week.

Table K.2: Three-week-long testing schedule

Date	Task	Personnel and location
Week 1 (Mar 23-29)	Prototype assembly, Arduino code, motor calibration, strip preparation	All team members at Myhal MyFab
Week 2 (Mar 30-Apr 5)	Calibration runs, tune if needed, testing runs.	Operator/Recorder at Myhal MyFab
Week 3 (Apr 6-8)	Complete data analysis, review video footage, compile final results	Operator/Recorder at Myhal MyFab

Appendix K.5: Success Criteria

The proposed rotary paper trimmer design will be considered successful if testing demonstrates that all three interdependent objectives are met together. Table K.3 details the success criteria for each objective.

Table K.3: Success Criteria by Objective

Objective	Metric	Target	Measurement Method
O.1 Human Interaction	Minutes spent interacting with the machine per day	< 60 minutes total (loading + interventions)	Sum of all loading times and intervention times recorded via stopwatch; projected to 8000 packages/day
O.2 Production Rate	Strips cut per hour	≥ 1750 strips per hour	Calculated from total run time and total packages processed; adjusted for cut

			interval
O.3 Success Rate	Undamaged packages per 1000 packages	≥ 99.9%	Visual inspection of all packages; failures categorized as misaligned cuts, torn packages, or jams

If the prototype barely misses one criterion but exceeds in others, the team will make note of this gap and make design modifications or future modifications for the design.

Appendix K.6: Limitations and Assumptions

The following limitations and assumptions apply to this test plan.

Limitations:

1. Limited samples: Only ~200 samples were available for testing; the majority of data will be collected using similar materials, which may not perfectly replicate the properties of the pouches
2. Visual inspection constraints: Without magnification tools or scales, minor damages to packages may go undetected, overstating the success rate
3. Controlled environment: Testing will be done in a university workspace, rather than an active hospital pharmacy. Results may not account for real-world interruptions or integration challenges.
4. Limited run count: With only 9 total runs, the sample size may not be large enough to capture all potential failure modes in performance

Assumptions:

1. Simulated paper strips (110-112 μm thickness) are reasonably close to the actual thickness, as both materials fall within similar thickness ranges
2. Results from approximately 1000 test packages can be reasonably extrapolated to a full 8000 packages
3. The two-person testing operation provides more reliable data, as tasks are distributed strictly
4. Visual inspection is sufficient to detect failures that would compromise medication integrity in a real pharmacy setting

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10.0 Appendices

Appendix A: Interest Holders Brainstorm

Table A1 shows the list of all interest holders identified through an Impact-Influence Matrix (Appendix B). This matrix helped categorize stakeholders based on how much they affect the system and how much the system affects them. The resulting list provides a comprehensive view of everyone with a meaningful role in or connection to the dispensing system.

Table A1. List of all interest holders.

#	Interest Holder
1	**Pharmacy Technicians**
2	Pharmacy Managers
3	**Biomedical Engineering Department**
4	Ergonomic Specialists
5	Canadian Union of Public Employees (CUPE)
6	BD/ Parata (ATP Vendors)
7	**Pharmacy Technicians**
8	Institute for Safe Medication Practices (ISMP)
9	Toronto General/ Sick Kids/ Princess Margaret Hospitals
10	Supply chain and inventory management staff
11	Patients
12	Cleaning Staff
13	Facility maintenance staff
14	Pharmaceutical Drugs Directorate (PDD)
15	Medical Devices Directorate (MDD)
16	Health Products and Food Branch (HPFB)
17	Assurance & Compliance Office
18	**Sinai Health**
19	Healthcare Insurance Providers
20	Medical Device Reprocessing (MDR) Technicians

Appendix B: Interest Holders Ranking

Table B1 ranks the interest holders from Appendix A using an Impact-Influence Matrix to evaluate each group's role in the project. Interest holders were positioned based on how strongly the system affects them and how much authority they have to shape its design and operation. The resulting categories clarify which groups must be prioritized during decision-making and which require ongoing consideration throughout the project.

Table B1. Analysis of the interest holders from Table A1.

High Impact, Low Influence 11, 12, 13	**High Impact, High Influence** 1, 3, 7, 18

Patients (11), Cleaning staff (12), and Facility maintenance staff (13) are placed here because while they are significantly impacted by the device's operations, they have little formal authority to influence its design and implementation. Patients are the beneficiaries of the design, as the design's reliability directly influences medication safety and delivery speeds. With the design planned across multiple wings, cleaning staff significantly impacted by the design's ease to clean and procedures. Similarly, the maintenance staff are responsible for the design's upkeep, where a poorly serviceable design could increase workload and decrease operational uptime. These interest holders share a high impact from the design due to their daily interactions, and their low influence results from not being included directly in the design process.	Pharmacy technicians (1), Biomedical engineering department (2), Pharmacy technicians (7), Sinai Health (18) each are essential to shaping the final design with their opinions. Biomedical engineers are required to maintain and upkeep the design, ensuring it is safely operational. Pharmacy technicians are the primary users of the device and benefit directly from the improvement in workflow the machine provides. Lastly, Sinai Health defines the project's alignment with their mission statement and organizes the possible large-scale distribution of the machine across its hospitals.
Low Impact, Low Influence 4, 5, 10, 15, 19, 20	**Low Impact, High Influence** 2, 6, 8, 9, 14, 16, 17
Ergonomic specialists (4), CUPE (5), Supply chain staff (10), MDD (15), Insurance providers (19), and MDR technicians (20), have limited influence over the implementation or design process. They act as advisors, upholding their regulations and protocols, however, are not specific to this design, and in most cases, do not apply to our scope. Their needs only become relevant in later stages, such as commercialization or implementation with other medical machines, resulting in these interest holders in this quadrant for the current project phase.	Pharmacy manager (2), BD/ Parata (6), ISMP (8), other hospitals (9), and Regulatory/ compliance offices (14, 16, 17) hold significant influence to enable or constrain the project but are less affected by it in the daily operations. The vendor (6) controls the compatibility with existing packages and ATP systems. ISMP (8) and other regulatory groups set regulations around the design. Nearby hospitals (9) represent the similar workflows the design will have to handle if it is to be adopted in the future, influencing scalability. The various internal offices (14, 16, 17) ensure compliance with Sinai Health policy compliance. While the daily operations of the machine do not depend on this device, their standards and relations set significant bounds on the design, giving them strong

	influence despite little direct impact.

Appendix C: The Black Box Method

This appendix shows the system used to define the design's functions, the Black Box Method (Table C1.). By focusing only on observable inputs and outputs, the method clarifies what the machine must achieve without specifying internal mechanisms. This approach helps establish the essential functional requirements that guide design work.

Table C1. Black box method to create functions

Input		Output
Mass: - Strips of pills enter the machine		Mass: - Individual packages of pills (strips of 2s, 5s, 10s)
Information: - Strips of pills with information on barcode goes into the machine		Information: - Strips of pills with information on barcode comes out of the machine (no change)
Energy: - N/A		Energy: - N/A

Appendix D: The Functional Basis

The functional basis identifies the core function of the system using a specific framework. For this project, the functional basis for this project would be Separate Mass, reflecting the device's role in dividing continuous pill strips into smaller packaged units. This classification established the foundation for defining the system's mechanical and operational requirements.

Appendix E: Objectives

This appendix outlines the design objectives generated in Table E1, which shows the full brainstorming list. These objectives were then prioritized through pairwise comparison shown in Table E2, establishing their relative importance for guiding the system's development.

Table E1. Objective Brainstorming List

Objectives Brainstorm
Does not require software updates (complexity) Less maintenance (money cost and amount of time per month)

Can load an entire strip per feed (e.g., >=100 packs per feed) Compact in size (Measured in m^3) Ideally works pill-size dependent Should be transparent to allow visual inspection Should ideally use a tearing motion Should work continuously for 2-5 hours

Table E2. Pairwise Comparison of the Objectives

	Success Rate	Cost	Human Interaction	Loading capacity	Allow visual inspection	Require minimal maintenance	Dimensions	Total Production	**Score**
Success Rate	X	1	0	1	1	1	1	0	**5**
Cost	0	X	0	0	0	0	0	0	**0**
Human Interaction	1	1	X	1	1	1	1	1	**7**
Loading Capacity	0	1	0	X	1	1	1	0	**4**
Allow visual inspection	1	0	0	0	X	0	1	0	**2**
Require minimal maintenance	0	1	0	0	1	X	1	0	**3**
Dimensions	0	1	0	0	0	0	X	0	**1**
Total Production	1	1	0	1	1	1	1	X	**6**

Appendix F: Constraints

This appendix lists the design constraints identified during the project, as summarized in Table F1, which captures the full constraint brainstorming output. These constraints define non-negotiable requirements the design must meet to ensure safety, functionality, and compatibility with existing workflows.

Table F1. Group Constraint List

Constraint Brainstorm

No blade/ exposed blade usage (Proly Constraint)
From ATP machine able to separate into strips of 1, 2, 5, 10 (constraint)
Eliminate tearing motion required by staff
Handle different pill sizes
Temperature change <= few degrees
Emergency stop function
Error rate (1 in 1000 at least)

Appendix G: Idea Generation Process

Appendix G documents the full development of means used to generate design concepts. Section G1 contains each team member's individual brainstorming, created using timed-thinking, blue-sky thinking, benchmarking, and in some cases early morph chart based solutions. Section G2 presents the combined list of means organized by function, followed by Section G3, which consolidates these entries and highlights duplicates in blue. Finally, Section G4 includes the morph chart constructed from the consolidated list, which was used to generate the team's set of 69 design ideas.

G.1: Individual Brainstorming

Name	Brainstorming
Alex	1. Robotic arms that tear bags based on a vision system 2. Abrasive discs and a roller for transporting packages 3. Regular guillotine and tracks: as the packages move through, a guillotine cuts connections 4. Have two clamps that grip the bags and tear, paired with a vision system 5. Any solution that uses low temperature to make the plastic bags brittle to assist in separation 6. Still track/conveyor belt, but have saw-tooth blades for clean separation (either punch or slice back and forth) 7. Use a roller conveyor and highly pressurized water 8. Heated knife that more easily cuts through bags. (potentially wire, low melting point of plastics, need a specific type or Tm determined) 9. Ultrasonic blades? 10. Conveyor belt and high-energy laser 11. Torque
Alba	Separation Mechanics: 1. Saw 2. X-acto Knife

	3. Laser Cutter 4. Punching/Stamping 5. Waterjets 6. Hot Knife 7. Freezing (causing brittleness) 8. Robot Arm Movement: 1. Robotic Arm 2. Carry by Hand 3. Use gravity to unroll (on a slant) 4. Conveyor Belt 5. Rollers 6. Robot Car 7. Vacuum 8. Floating on Water Keeping in Place: 1. Clamps (on edges of strips) 2. Suction 3. BluTack 4. Robotic Arm 5. Carry by Hand 6. Rollers (on top) 7. Robot Car 8. Vacuum 9. Floated on Water 10. Metal Plates (on edges) Morph Chart:

	Means	Means	Means	Means	Means	Means
Separate tablet packages	Saw	X-acto Knife	Punching /Stamping	Robot Arm	Laser Cutter	Hot Knife
Advance the tablet strip within the machine	Robot Arm	Gravity	Conveyor belt	Rollers	Vacuum	

Boen

Sensing & Alignment

1. Color / Contrast Sensor - Detects the printed line or varying light transmission between the pouch and the pill to trigger the cutting mechanism.
2. Mechanical Micro-switch - Uses a small roller arm that physically drops into the perforated groove between pouches to activate the cut.
3. Stepper Motor Dead-Reckoning - Feeds the strip by a precise, pre-programmed distance (e.g., 50mm per pouch) without relying on external optical sensors.
4. V-shaped Guide Funnel - A purely passive mechanical guide that flattens and aligns the twisted strip before it enters the cutting zone.
5. Spring-loaded Tensioners - Applies constant pulling force at both ends of the cutting zone to keep the plastic strip taut and prevent wrinkling.

Pill Repositioning

1. Gravity Tilt Orientation - Angles the entire feeding track (e.g., at 45 degrees) so pills naturally fall to the bottom edge of the pouches, away from the cutting line.
2. Soft Rubber Wiper - A flexible paddle that sweeps across the pouch to gently push the pill away from the perforation before the blade descends.
3. Foam Squeeze Rollers - Soft, compressible rollers that grip the pouch and naturally displace the hard pill to the side without crushing it.
4. Nylon Brush Sweeper - A row of soft bristles that brush over the strip to flatten the packaging and gently push pills aside.

Cutting Mechanisms

1. Rotary Paper Trimmer - A sharp circular blade that slides laterally across a fixed track to slice the plastic cleanly.

	2. Serrated Guillotine Blade - A downward-pressing blade with jagged teeth to easily pierce and shear the smooth plastic packaging. 3. Differential Speed Tearing - Uses two sets of rollers where the secondary rollers spin faster, snapping the strip along the pre-existing perforations via tension. 4. V-shaped Punch Blade - A blade shaped like a 'V' that concentrates the cutting force at the center of the strip for a cleaner, non-slip cut. Transportation / Movement 1. Dual Rubber Tank Treads - Two parallel belts that sandwich the strip from top and bottom to provide high friction and prevent slipping. 2. Edge Pinch Rollers - Small rollers that only grip the flat, heat-sealed edges of the strip, ensuring the central pill pocket is never compressed. 3. Intermittent Geneva Drive - A mechanical gear system that translates continuous rotation into intermittent stop-and-go motion, allowing a pause for the cut. Output / Collection 4. Motorized Diverter Flap - A mechanical gate that swings left or right to sort single pouches into one bin and larger strips (e.g., 5s or 10s) into another. 5. Slanted Collection Chute - A simple gravity-fed ramp that guides the cut packages directly into standard hospital storage bins. 6. Drop-count Photoelectric Sensor - An infrared beam sensor placed at the exit that counts the falling packages to track batch sizes.
James	Cutting Mechanisms 1. Guillotine-style blades - drop from above at a specific angle 2. Rotary cutting blade - spinning wheel with blades attached at set intervals, spins to cut, adjustable speed to accommodate batch sizes 3. Scissor mechanism - two blades attached closely to act like scissors 4. Wire cutters - heated wire which burns through the plastic quickly 5. Laser cutter - small laser burns through parts of the packaging with improved precision compared to wire cutters 6. Waterjet cutter - high-pressure water to split the packages Transportation mechanisms 7. Friction rollers - rubber studs on the roller 8. Toothed rollers - teeth grab on the side of the bag 9. Vacuum conveyor - suction holds the strip to the conveyor 10. Pinching wheel - two wheels pinch the top and bottom of the packaging, “spits” the roll out 11. Lever pusher - when the batch is cut, it falls and hits a lever, which allows the next batch to advance

12. Spaced perforation rollers - rollers with set teeth at the perforations of the packages, teeth eat into the perforations and advance it forward, can also be a cutting mechanism
13. Tilted drop-chute - the machine is tilted at a specific angle, cut strips fall

Pill repositioning mechanisms

14. Vibrating, tilted track - vibrates and tilts the pill near the preferred edge of the package
15. Compression rollers - gradually roll/ guide the tablet to the edge of the roller
16. Angled rails - forces the pills onto an edge
17. Camera detection - detects the position of the pill, adjusts the advance of the strip to avoid pill damage

Pill output/storage

18. Stacking tray - cut strips fall into a tray with adjustable sizes to accommodate batch cut size

SCAMPER

19. Original: Banknote counting/ sorting machine
    a. Modify/ Separate: Lever arm with heated wire, cuts the pouch at given intervals
    b. Adapt/ Advance: pinching rollers on the sides
    c. Substitute/ Reposition: pills automatically drop down due to gravity
    d. Combine/ Output: output into stacks
20. Original: Label printer
    a. Modify/ Separate: modified the tear bar into a moving cutter wheel/ blade that swipes whenever a certain batch length is achieved
    b. Adapt/ Advance: Adapted the current label feeding mechanism to accommodate our dimensions
    c. Substitute/ Reposition: Combine the feeding track with an angled pill pusher/ guide
    d. Put to another use/ Output: repurpose the tray where labels fall into for an organized pouch strip stack
21. Original: Cricut machine (Vinyl cutter)
    a. Substitute/ Separate: Modify the blade to accommodate cutting plastic
    b. Adapt/ Advance: Adapt the feeding mechanism to handle long pill strips
    c. Combine/ Reposition: add a second tool which pushes the pill to the sides of the machine automatically using cameras and AI
    d. Modify/ Output: repurpose the output to have a container to hold pill strips
22. Original: Heat sealer for food
    a. Substitute/ Separate: substitute the heating bar with wire elements that can cut through plastic, include a vent to vacuum up cutting odours
    b. Adapt/ Advance: Adapted the existing conveyor system with tracks that grip the packages

	c. Combine/ Reposition: Combine compressing rollers with feed guides to reposition pills d. Modify/ Output: Modified the cooling section to include an output tray and to cool the cut sections, preventing melted plastic 23. Original: 3D printer a. Modify/ Separate: substitute the filament extruder with a low-powered laser module for cutting b. Substitute/ Advance: Substitute the printing bed with conveyor grips to hold the strip in place during cutting c. Combine/ Reposition: Repositions the laser head with the cameras d. Adapt/ Output: Adapt the front ledge of the printer to have an output basket
Maria	S ● Substitute fixed blades with interchangeable cutting blades. ● Substitute standard conveyor belt with roller feed system. ● Substitute manual loading with gravity-fed system. C ● Combine feeding and alignment guides into a single adjustable track. ● Combine counting mechanism + length detection. ● Combine clamping mechanism + cutter for more accurate separation. A ● Add adjustable guide rails ● Add an emergency stop and safety guard. ● Add predictive maintenance sensors. M ● Modify blade angle for cleaner cuts. ● Modify feeding angle to reduce slippage. ● Modify separation force to avoid pill damage. ● Change from cut-based separation → perforation tearing mechanism. P ● Use paper trimmer mechanism principles. E ● Eliminate multiple conveyors — use a single feed path. R ● Rearrange process to measure → clamp → cut → release → stack. ● Rotate cutting mechanism instead of moving the strip. Blue-Sky Thinking ● Vibration to push pills to the side ● Compression roller ● Angled conveyor surface

	• Air pulse
Nolan	Separate tablet packages 1. Tearing 2. Hinged blade 3. Stationary blade (table saw) 4. Laser cutter Advance the tablet strip within the machine 1. Wheels 2. Conveyor belt Reposition tablets within the package pouch 1. Track 2. Pushing arm Output tablet strips 1. Slide

G.2: Means separated by functions

PF. 1	Separate tablet packages

1. Tearing
2. Hinged blade
3. Stationary blade (table saw)
4. Laser cutter
5. Guillotine-style blades - drop from above at a specific angle
6. Rotary cutting blade - spinning wheel with blades attached at set intervals, spins to cut, adjustable speed to accommodate batch sizes
7. Scissor mechanism - two blades attached closely to act like scissors
8. Wire cutters - heated wire which burns through the plastic quickly
9. Laser cutter - small laser burns through parts of the packaging with improved precision compared to wire cutters
10. Waterjet cutter - high-pressure water to split the packages
11. Saw
12. X-acto Knife
13. Laser Cutter
14. Punching/Stamping
15. Waterjets
16. Hot Knife
17. Freezing (causing brittleness)

18. Robot Arm
19. Offshore cutting task to Asia, China or the Philippines
20. Tearing bags
21. Machete
22. Table saw
23. Pliers
24. Heat (flame torch)
25. Interchangeable cutting blades (circular motion) instead of one blade that goes up and down
26. Adjustable blade angles
27. Tear pills apart instead of cut
28. Paper trimming mechanism
29. High speed rotary cutter
30. Abrasive discs
31. Rotary Paper Trimmer
32. Serrated Guillotine Blade
33. Uses low temperature to make the plastic bags brittle to assist separation
34. Saw-tooth blades for clean separation (either punch or slice back and forth)
35. Differential Speed Tearing
36. Ultrasonic blades (vibrating blade)
37. V-shaped Punch Blade

SF. 1	Advance the tablet strip within the machine

1. Wheels
2. Conveyor belt
3. Robotic Arm
4. Carry by Hand
5. Use gravity to unroll (on a slant)
6. Conveyor Belt
7. Rollers
8. Robot Car
9. Vacuum
10. Floating on Water
11. Friction rollers - rubber studs on the roller
12. Toothed rollers - teeth grab on the side of the bag
13. Vacuum conveyor - suction holds the strip to the conveyor
14. Pinching wheel - two wheels pinch the top and bottom of the packaging, “spits” the roll out
15. Lever pusher - when the batch is cut, it falls and hits a lever, which allows the next batch to advance
16. Spaced perforation rollers - rollers with set teeth at the perforations of the packages, teeth eat into the perforations and advance it forward, can also be a cutting mechanism

17. Tilted drop-chute - the machine is tilted at a specific angle, cut strips fall 18. Roller feed system 19. Gravity feed system 20. Single track 21. Guide rails 22. and a roller for transporting packages 23. V-groove track	
SF. 2	Reposition tablets within the package pouch
1. Track 2. Pushing arm 3. Vibrating, tilted track - vibrates and tilts the pill near the preferred edge of the package 4. Compression rollers - gradually roll/ guide the tablet to the edge of the roller 5. Angled rails - forces the pills onto an edge 6. Camera detection - detects the position of the pill, adjusts the advance of the strip to avoid pill damage 7. Clamps (on edges of strips) 8. Robotic Arm 9. Carry by Hand 10. Rollers (on top) 11. Robot Car 12. Vacuum 13. Floated on Water 14. Metal Plates (on edges) 15. Maintenance sensors 16. Grip on track to avoid slippage 17. Mechanism to push pills to the side 18. Vibration counterbelt to push the pill to one side 19. Angled track 20. Air pulse to encourage the pill to move to the side 21. Compression roller that will push the pill to one side	
SF. 3	Output tablet strips
1. Slide 2. Stacking tray - cut strips fall into a tray with adjustable sizes to accommodate batch cut size	

G.3: Means consolidation

Below is a list of complete and consolidated ideas, with similar or duplicate ideas removed.
Duplicates are highlighted with this colour

PF. 1	Separate tablet packages

1. Hinged blade
2. Stationary blade (table saw)
3. Laser cutter
4. Guillotine-style blades - drop from above at a specific angle
5. Rotary cutting blade - spinning wheel with blades attached at set intervals, spins to cut, adjustable speed to accommodate batch sizes
6. Scissor mechanism - two blades attached closely to act like scissors
7. Wire cutters - heated wire which burns through the plastic quickly
8. Laser cutter - small laser burns through parts of the packaging with improved precision compared to wire cutters
9. Waterjet cutter - high-pressure water to split the packages
10. Saw
11. X-acto Knife
12. Laser Cutter
13. Punching/Stamping
14. Waterjets
15. Hot Knife
16. Freezing (causing brittleness)
17. Robot Arm
18. Tearing bags
19. Machete
20. Table saw
21. Pliers
22. Heat (flame torch)
23. Interchangeable cutting blades (circular motion) instead of one blade that goes up and down
24. Adjustable blade angles
25. Tear pills apart instead of cutting
26. Paper trimming mechanism
27. High-speed rotary cutter
28. Abrasive discs
29. Rotary Paper Trimmer
30. Serrated Guillotine Blade
31. Uses low temperature to make the plastic bags brittle to assist in separation
32. Saw-tooth blades for clean separation (either punch or slice back and forth)
33. Differential Speed Tearing
34. Ultrasonic blades
35. V-shaped Punch Blade

SF. 1	Advance the tablet strip within the machine

1. Wheels
2. Conveyor belt
3. Robotic Arm
4. Carry by Hand
5. Use gravity to unroll (on a slant)
6. Conveyor Belt
7. Rollers
8. Robot Car
9. Vacuum
10. Floating on Water
11. Friction rollers - rubber studs on the roller
12. Toothed rollers - teeth grab on the side of the bag
13. Vacuum conveyor - suction holds the strip to the conveyor
14. Pinching wheel - two wheels pinch the top and bottom of the packaging, “spits” the roll out
15. Lever pusher - when the batch is cut, it falls and hits a lever, which allows the next batch to advance
16. Spaced perforation rollers - rollers with set teeth at the perforations of the packages, teeth eat into the perforations and advance it forward, can also be a cutting mechanism
17. Tilted drop-chute - the machine is tilted at a specific angle, cut strips fall
18. Roller feed system
19. Gravity feed system
20. Single track
21. Guide rails and a roller for transporting packages
22. V-groove track

SF. 2	Reposition tablets within the package pouch

1. Track
2. Pushing arm
3. Vibrating, tilted track - vibrates and tilts the pill near the preferred edge of the package
4. Compression rollers - gradually roll/ guide the tablet to the edge of the roller
5. Angled rails - forces the pills onto an edge
6. Camera detection - detects the position of the pill, adjusts the advance of the strip to avoid pill damage
7. Clamps (on edges of strips)
8. Robotic Arm
9. Carry by Hand
10. Rollers (on top)
11. Robot Car

12. Vacuum 13. Floated on Water 14. Metal Plates (on edges) 15. Maintenance sensors 16. Grip on track to avoid slippage 17. Vibration counterbelt to push the pill to one side 18. Angled track 19. Air pulse to encourage the pill to move to the side 20. A compression roller that will push the pill to one side	
SF. 3	Output tablet strips
1. Slide 2. Stacking tray - cut strips fall into a tray with adjustable sizes to accommodate batch cut size	

G.4: Idea formulation

Linked below is the morph chart that was used to create the list of ideas found in H1 completed via Google Sheets.
https://docs.google.com/spreadsheets/d/10W6SA1ihasgjWV9eTHYyPtGM8L0IM6o1nsa9F4rWy3o/edit?usp=sharing

Appendix H: Idea Selection Process

Appendix H documents the full progression from idea generation to concept selection, moving from an initial set of 69 ideas through consolidation, feasibility evaluation, multivoting, and final design selection.
** After the selection process was completed, our three proposed alternative designs were labelled with their names to improve the readability of the idea selection process.*

H.1: Full List

This section presents the complete list of 69 ideas generated using the morph chart in Appendix G4. Duplicate concepts are highlighted in blue to make the later consolidation process more efficient and transparent. This section captures the breadth of the team's early design exploration before any filtering or evaluation.

1. **Rail-Aligned Rotary Vacuum Cutter:** A vacuum conveyor holds the strip flat while a rotary cutting blade trims through the plastic. Angled rails gradually push the pill to a safe edge, and the cut piece drops into an adjustable stacking tray.

2. **SmartLaser Pinch-Wheel Separator:** Pinch wheels move the strip forward while a camera detects pill position. A laser cuts around it, and the strip falls into a collection tray.
3. **Vision-Laser MicroCut Conveyor:** A conveyor feeds the strip under a camera-guided laser that adjusts the cut in real time. A slide directs the finished pieces into a bin.
4. **HydroJet Lever-Feed Splitter:** A lever pusher advances the strip beneath a waterjet. A small air pulse pushes the pill aside, and the cut strip falls into a tray.
5. **Ultrasonic Rail-Guided Cutter:** Ultrasonic blades cut the advancing strip on a conveyor. Angled rails push the pill away before the cut, and pieces fall into a bin.
6. **High-Frequency Ultrasonic Slide Cutter:** A conveyor brings strips beneath an ultrasonic tool that vibrates rapidly to slice the film. Plastic rails shift the pill aside, and a slide collects the pieces.
7. **Vacuum-Stabilized Rotating Blade Track:** A rotating blade slices the strip as a vacuum suction panel stabilizes movement. Two angled guide rails progressively shift pills aside before cutting and a slide collects the output.
8. **Lever-Driven Waterjet Slicer:** A high-pressure waterjet cuts strips fed by a lever mechanism, with air jets aligning the pill before slicing. Output slides down a chute.
9. **Robo-Punch Guidance System:** A robotic arm positions each pouch under a stamping punch. A vibrating tilted track ensures the pill slides toward the safe edge before cutting.
10. **Automated Punch-Robot Feeder:** A robot places the strip under a stamping die while micro-vibrations reposition pills. Segments drop into a stacking tray.
11. **Heated Wire Roller Advance Cutter (ThermoCut*):** Studded rollers push the strip beneath a heated wire that melts through the packaging. A slide guides the pieces down.
12. **Thermal Wire Melt-Line Roller Cutter:** Friction rollers feed the plastic strip into a heated cutting wire, and compression rollers push the pill toward an edge. Output pieces fall into a tray.
13. **V-Blade Vacuum Punch Station:** A vacuum belt locks the strip to the surface while a V-shaped punch slices downward. Air pulses reposition the pill.
14. **Abrasive Disc Pinch-Feed Cutter:** A rotating abrasive disc cuts the plastic while a pinch wheel feeds the strip. Compression rollers shift the pill aside.
15. **Disc-Wheel Abrasion Slicer:** A small abrasive wheel trims the strip fed by friction rollers. The pill is guided away by angled rails.

16. **Cryo-Crack Belt Separator:** A freezing nozzle chills the plastic until brittle, and a conveyor snaps it along a guide rail.
17. **Freeze-Break Rail Channel Cutter:** The strip is cooled along a rail until it fractures cleanly. A chute directs the pieces downward.
18. **Vacuum-Guided V-Punch Separator:** The strip is held by vacuum while a V-blade removes a small segment. Air jets keep the pill away from the cutting zone.
19. **Tear-Roller Differential Cutter:** Friction rollers rotate at offset speeds to tear the strip along a seam. Grip rails guide the pill aside.
20. **Rotary Trim-Rail Alignment System:** A high-speed rotary blade cuts strips while angled rails push the pill away. A slide captures output.
21. **High-Speed Rotary Rail Cutter:** A spinning cutting wheel slices the strip fed by rollers; rails shift the pill aside.
22. **Pizza-Wheel Foam Roller Cutter:** Foam rollers push the pill aside as a pizza-wheel blade slices through the packaging.
23. **Rotary Pizza-Blade Pill-Safe Trimmer:** A rolling pizza-style cutter trims plastic strips guided by foam wheels.
24. **Saw-Tooth Pinch Roller Station:** A saw-tooth blade slices strips fed by pinch rollers, with compression rollers repositioning the pill.
25. **Differential Speed Tear Station:** Different roller speeds pull the strip apart along perforations while rails align the pill.
26. **Serrated Saw-Feed Cutter:** A serrated blade cuts plastic fed by pinch rollers; rails align the pill away.
27. **Gravity Slide Punch Separator:** A slanted gravity-slide moves the strip into a punching blade. Angled rails position the pill beforehand.
28. **Vacuum-Scissor Rail Cutter:** A vacuum conveyor holds the strip while scissors cut through the packaging. Compression rollers reposition pills.
29. **Cooling Blade Gravity Dropper:** A cooling air jet chills the strip, and a vertical blade slices the weakened seam as gravity pulls it downward.
30. **Frost-Edge Gravity Cutter:** Cold air makes the plastic brittle before a static blade slices it.
31. **Dual-Roll Guillotine Feed System:** Rollers feed the strip into a guillotine blade while compression rollers shift pills away.

32. **Clamp-Stabilized Rotary Trim Unit:** Edge clamps hold the strip as a paper-trimmer style rotary cutter slices it.
33. **Trimmer-Clamp Edge-Feed System:** Clamps secure the strip as a rotary trimmer advances along the cutline.
34. **Vacuum-Corner Robotic Cutter:** A robotic arm places strips against a fixed cutting blade while a vacuum draws pills to a safe corner.
35. **Robo-Blade Vacuum Positioning Station:**A robot positions the strip and vacuum pulls pills aside before cutting.
36. **Laser-Camera Pinch Rail Station:** A laser cuts strips advanced by pinch wheels, guided by camera detection and angled rails.
37. **Slanted Punch Gravity-Feed Channel:** Strips slide down a slope into a punch blade while rails reposition the pill.
38. **Camera-Laser Rail-Guide Cutter:** A camera maps pill location for a laser cut as rails force the pill away.
39. **Roller-Advance Guillotine Assembly:** Rollers reposition pills and feed the strip directly under a guillotine.
40. **Scissor-Vacuum Alignment Cutter:** A scissor mechanism cuts the strip stabilized by vacuum suction and shifted by rails.
41. **Vacuum driven rail aligned rotary cutter:** A rotary cutting blade slices strips as they advance on a vacuum conveyor, while angled rails reposition pills against pocket edges, and finished pieces fall into an adjustable stacking tray.
42. **Guillotine Friction Compression Rollers:** A guillotine blade drops to cut strips fed by friction rollers, while compression rollers gradually guide tablets to the edge, and cut strips slide down a slide into a collection area.
43. **Laser Cutter with Pinching Conveyor:** A laser cutter burns through packaging as a pinching conveyor advances the strip, while camera detection repositions the cut to avoid pill damage, and pieces drop into a stacking tray.
44. **Lever and Air Driven Water Jet:** A waterjet cutter splits packages fed by a lever pusher mechanism, while air pulses encourage pills to move to the side, and finished strips fall onto a slide.
45. **Automated Robotic Punching Station:** A punching/stamping station cuts pouches as a robotic arm places them into position, while vibrating tilted track repositions pills, and output collects in an adjustable stacking tray.

46. **Ultrasonic Blade Fed by Gravity and Foam Roller:** Ultrasonic blades cut strips transported by gravity, while foam rollers force pills onto a safe edge, and pieces slide down a slide into a bin.
47. **Rotary Paper Trimmer with Rollers and Clamps for Positioning:** A rotary paper trimmer cuts strips fed by rollers, while clamps on edges reposition the strip for accuracy, and finished pieces fall into a stacking tray.
48. **Saw with Gravity and Grippy Track:** A saw cuts strips as they advance via gravity on a slant, while grip on track prevents slippage for repositioning, and output slides down a slide.
49. **Heated Wire with Studded Rollers:** A heated wire cutter burns through plastic as studded rollers advance the strip, while camera detection repositions the cut, and pieces collect in a stacking tray.
50. **V-Shaped Blade with Vacuum Conveyor and Air Pulse:** A V-shaped punch blade cuts strips held by a vacuum conveyor, while air pulse repositions pills, and output drops into an adjustable stacking tray.
51. **Abrasive Disc with Pinching Wheel and Compression Rollers:** An abrasive disc cuts strips fed by a pinching wheel, while compression rollers guide tablets to the edge, and pieces slide down a slide.
52. **Freeze with Conveyor Belt and Angled Rails:** A freezing mechanism embrittles and breaks plastic on a conveyor belt, while angled rails reposition the strip, and brittle pieces break into a stacking tray.
53. **Tearing with Friction Rollers:** Differential speed tearing separates strips advanced by friction rollers, while grip on track repositions the material, and torn pieces fall onto a slide.
54. **Angled Rotary Cutter:** A high-speed rotary cutter slices strips fed by rollers, while angled rails force pills to the edge, and output slides down a slide.
55. **Rotary Cutter and Wheels (PharmaSlice*):** A pizza cutter slices strips fed by roller wheels (with foam to accommodate for different pill sizes), while the wheels forces pills to the edge, and outputs into a basket
56. **Saw-Tooth Blade Cutter with Pinching Wheel Rollers:** A saw-tooth blade cuts strips fed by a pinching wheel, while compression rollers guide tablets, and finished pieces collect in a stacking tray.

57. **Punching with Gravity Feed and Angled Rails for Pill Position:** A punching/stamping station cuts strips fed by gravity on a slant, while angled rails reposition the strip, and pieces drop into a stacking tray.
58. **Scissor Vacuum with Compression Rollers:** A scissor mechanism cuts strips on a vacuum conveyor, while compression rollers guide tablets to the edge, and finished pieces drop into a stacking tray.
59. **Gravity Cooling Vertical Blade:** The strip feeds downward on a slanted path using gravity. Cooling makes the package more brittle, while metal edge plates keep it aligned. Once weakened, the package is separated via a vertical blade and the finished strip travels down the slide.
60. **Heated Wire Gravity Slide:** A hot wire cuts strips, speed fluctuating depending on the number of packages, and uses gravity to slide down the path.
61. **Rollers and Guillotine Blade:** Rollers move the strip through the system. Compression rollers push the pill away from the cutting zone. A guillotine blade cuts at a controlled angle, and the separated segment exits via a slide.
62. **Scissors with Edge Clamps and Conveyor Belt:** The conveyor carries the strip through the machine while edge clamps stabilize it. The scissor blades cut at the target location. The cut strips then move down a slide for collection.
63. **Robotic Arm:** A robotic arm that performs all means (the already existing solution but without any humans).
64. **Guillotine with Two Conveyor Belts and Foam Rollers:** Blade that cuts downwards with guardrails, made of two conveyor belts (the space in between is for the blade to pass through) and foam roller to push pills to the side
65. **Ultrasonic Blade with Friction Roller:** Friction rollers pull the strip forward while angled rails shift the pill toward a safer edge. Ultrasonic blades cut with reduced mechanical force, helping reduce deformation of the package. The finished pouch exits through a slide.
66. **Robotic Arm with Vacuum and Fixed Cutting Blade:** A robotic arm places and positions each strip against a fixed cutting blade. A vacuum mechanism moves the pills to a specific corner during the cut. The finished pieces are then deposited into a stacking tray.
67. **Pinching Wheel with Camera and Laser:** A pinching wheel moves the pouches strip forward in small increments. Camera detection identifies the pill's position and ensures

the cut avoids the tablet. A laser makes a precise cut, and the finished unit drops into the tray.

68. **Rotary Paper Trimmer with Angled Rails:** Perforation rollers advance the strip by engaging repeated package features. Angled rails encourage the pill toward one side before cutting. A rotary paper-trimmer style cutter then slices the strip, and the finished piece exits on a slide.
69. **Hinged Blade with Roller (SliceFlow*):** The conveyor brings the pouch strip into the cutting zone. A roller helps reposition the pill or the strip before cutting. The hinged blade swings through the material, and the cut section exits down a slide.

H.2: Consolidation

This section contains the combined and cleaned list of ideas after removing duplicates and merging similar concepts. This consolidation step ensures that only unique, clearly differentiated ideas progress to evaluation. This resulting list forms the basis for the feasibility check that follows.

1. Rail-Aligned Rotary Vacuum Cutter
2. SmartLaser Pinch-Wheel Separator
4. HydroJet Lever-Feed Splitter
5. Ultrasonic Rail-Guided Cutter
9. Robo-Punch Guidance System
11. Heated Wire Roller Advance Cutter (ThermoCut*)
13. V-Blade Vacuum Punch Station
14. Abrasive Disc Pinch-Feed Cutter
16. Cryo-Crack Belt Separator
19. Tear-Roller Differential Cutter
20. Rotary Trim-Rail Alignment System
22. Pizza-Wheel Foam Roller Cutter
24. Saw-Tooth Pinch Roller Station
27. Gravity Slide Punch Separator
28. Vacuum-Scissor Rail Cutter
29. Cooling Blade Gravity Dropper
31. Dual-Roll Guillotine Feed System
32. Clamp-Stabalized Rotary Trim Unit
34. Vacuum-Corner Robotic Cutter
36. Laser-Camera Pinch Rail Station
41. Vacuum driven rail aligned rotary cutter
44. Lever and Air Driven Water Jet:
45. Automated Robotic Punching Station
46. Ultrasonic Blade Fed by Gravity and Foam Roller
47. Rotary Paper Trimmer with Rollers and Clamps for Positioning
48. Saw with Gravity and Grippy Track

49. Heated Wire with Studded Rollers
50. V-Shaped Blade with Vacuum Conveyor and Air Pulse
51. Abrasive Disc with Pinching Wheel and Compression Rollers
52. Freeze with Conveyor Belt and Angled Rails
53. Tearing with Friction Rollers
54. Angled Rotary Cutter
55. Rotary Cutter and Wheels (PharmaSlice*)
56. Saw-Tooth Blade Cutter with Pinching Wheel Rollers
57. Punching with Gravity Feed and Angled Rails for Pill Position
58. Scissor Vacuum with Compression Rollers
59. Gravity Cooling Vertical Blade
60. Heated Wire Gravity Slide
62. Scissors with Edge Clamps and Conveyor Belt
63. Robotic Arm
66. Robotic Arm with Vacuum and Fixed Cutting Blade
69. Hinged Blade with Roller (SliceFlow*)

H.3: Feasibility Check

This section evaluates each consolidated idea against the system's functions and constraints. Ideas that meet all requirements are highlighted in green, while those that fail one or more criteria appear in red and are removed from further consideration. This step ensures that only viable concepts continue to the multivoting stage.

Table H.3.1 The Feasibility Check Process

Idea	PF.1	SF.1	SF. 2	SF. 3	C. 1	C. 2	C. 3	C. 4	C. 5
1. Rail-Aligned Rotary Vacuum Cutter	Y	Y	Y	Y	Y	Y	Y	Y	Y
2. SmartLaser Pinch-Wheel Separator	Y	Y	Y	Y	Y	Y	Y	Y	N
4. HydroJet Lever-Feed Splitter	Y	Y	Y	Y	Y	N	Y	Y	Y
5. Ultrasonic Rail-Guided Cutter	Y	Y	Y	Y	Y	Y	Y	Y	Y
9. Robo-Punch Guidance System	Y	Y	Y	Y	Y	Y	Y	Y	N
11. Heated Wire Roller Advance Cutter (ThermoCut*)	Y	Y	Y	Y	Y	Y	Y	Y	Y
13. V-Blade Vacuum Punch Station	Y	Y	Y	Y	Y	Y	Y	Y	Y
14. Abrasive Disc Pinch-Feed Cutter	Y	Y	Y	Y	Y	Y	Y	Y	Y

16. Cryo-Crack Belt Separator	Y	Y	Y	Y	Y	N	Y	Y	N
19. Tear-Roller Differential Cutter	Y	Y	Y	Y	Y	Y	Y	Y	Y
20. Rotary Trim-Rail Alignment System	Y	Y	Y	Y	Y	Y	Y	Y	Y
22. Pizza-Wheel Foam Roller Cutter	Y	Y	Y	Y	Y	Y	Y	Y	Y
24. Saw-Tooth Pinch Roller Station	Y	Y	Y	Y	Y	Y	Y	Y	Y
27. Gravity Slide Punch Separator	Y	Y	Y	Y	Y	Y	Y	Y	Y
28. Vacuum-Scissor Rail Cutter	Y	Y	Y	Y	Y	Y	Y	Y	Y
29. Cooling Blade Gravity Dropper	Y	Y	Y	Y	Y	N	N	Y	N
31. Dual-Roll Guillotine Feed System	Y	Y	Y	Y	Y	Y	Y	Y	Y
32. Clamp-Stabalized Rotary Trim Unit	Y	Y	Y	Y	Y	Y	Y	Y	Y
34. Vacuum-Corner Robotic Cutter	Y	Y	Y	Y	Y	Y	Y	Y	N
36. Laser-Camera Pinch Rail Station	Y	Y	Y	Y	Y	Y	Y	Y	N
41. Vacuum driven rail aligned rotary cutter	Y	Y	Y	Y	Y	Y	Y	Y	N
44. Lever and Air Driven Water Jet:	Y	Y	Y	Y	Y	N	Y	Y	N
45. Automated Robotic Punching Station	Y	Y	Y	Y	Y	Y	Y	Y	N
46. Ultrasonic Blade Fed by Gravity and Foam Roller	Y	Y	Y	Y	Y	Y	Y	Y	Y
47. Rotary Paper Trimmer with Rollers and Clamps for Positioning	Y	Y	Y	Y	Y	Y	Y	Y	Y
48. Saw with Gravity and Grippy Track	Y	Y	Y	Y	Y	Y	N	Y	Y
49. Heated Wire with Studded Rollers	Y	Y	Y	Y	Y	Y	Y	Y	Y
50. V-Shaped Blade with Vacuum Conveyor and Air Pulse	Y	Y	Y	Y	Y	Y	Y	N	Y
51. Abrasive Disc with Pinching Wheel and Compression Rollers	Y	Y	Y	Y	Y	N	Y	Y	Y
52. Freeze with Conveyor Belt and Angled Rails	Y	Y	Y	Y	Y	Y	Y	Y	Y
53. Tearing with Friction Rollers	Y	Y	Y	Y	Y	N	Y	Y	N
54. Angled Rotary Cutter	Y	Y	Y	Y	Y	Y	Y	Y	Y

55. Rotary Cutter and Wheels (PharmaSlice*)	Y	Y	Y	Y	Y	Y	Y	Y	Y
56. Saw-Tooth Blade Cutter with Pinching Wheel Rollers	Y	Y	Y	Y	Y	N	N	Y	Y
57. Punching with Gravity Feed and Angled Rails for Pill Position	Y	Y	Y	Y	Y	N	Y	Y	Y
58. Scissor Vacuum with Compression Rollers	Y	Y	Y	Y	Y	Y	Y	Y	N
59. Gravity Cooling Vertical Blade	Y	Y	Y	Y	Y	Y	N	Y	N
60. Heated Wire Gravity Slide	Y	Y	Y	Y	Y	Y	Y	Y	Y
62. Scissors with Edge Clamps and Conveyor Belt	Y	Y	Y	Y	Y	Y	Y	Y	Y
63. Robotic Arm	Y	Y	Y	Y	Y	Y	Y	Y	N
66. Robotic Arm with Vacuum and Fixed Cutting Blade	Y	Y	Y	Y	Y	Y	Y	Y	N
69. Hinged Blade with Roller (SliceFlow*)	Y	Y	Y	Y	Y	Y	Y	Y	Y

H.4: Multivoting

This section shows the multivoting procedure used to narrow the feasible ideas down to a

Table H.4.1. The Multivoting Process

Round 1		**Round 2**		**Final 9 Ideas**
Idea	**Voting**	**Idea**	**Voting**	
1. Rail-Aligned Rotary Vacuum Cutter	II	1. Rail-Aligned Rotary Vacuum Cutter	I	
5. Ultrasonic Rail-Guided Cutter	II	5. Ultrasonic Rail-Guided Cutter	II	5. Ultrasonic Rail-Guided Cutter
11. Heated Wire Roller Advance Cutter (ThermoCut*)	I	11. Heated Wire Roller Advance Cutter (ThermoCut*)	III	11. Heated Wire Roller Advance Cutter (ThermoCut*)
13. V-Blade Vacuum Punch Station				
14. Abrasive Disc Pinch-Feed Cutter	II	14. Abrasive Disc Pinch-Feed Cutter	I	
19. Tear-Roller Differential Cutter				

20. Rotary Trim-Rail Alignment System				
22. Pizza-Wheel Foam Roller Cutter	II	22. Pizza-Wheel Foam Roller Cutter	II	22. Pizza-Wheel Foam Roller Cutter
24. Saw-Tooth Pinch Roller Station				
27. Gravity Slide Punch Separator	I	27. Gravity Slide Punch Separator	I	
28. Vacuum-Scissor Rail Cutter				
31. Dual-Roll Guillotine Feed System	II	31. Dual-Roll Guillotine Feed System	II	31. Dual-Roll Guillotine Feed System
32. Clamp-Stabalized Rotary Trim Unit	II	32. Clamp-Stabalized Rotary Trim Unit	II	32. Clamp-Stabalized Rotary Trim Unit
46. Ultrasonic Blade Fed by Gravity and Foam Roller	II	46. Ultrasonic Blade Fed by Gravity and Foam Roller	I	
47. Rotary Paper Trimmer with Rollers and Clamps for Positioning	II	47. Rotary Paper Trimmer with Rollers and Clamps for Positioning	II	47. Rotary Paper Trimmer with Rollers and Clamps for Positioning
49. Heated Wire with Studded Rollers	II	49. Heated Wire with Studded Rollers	I	
52. Freeze with Conveyor Belt and Angled Rails				
55. Rotary Cutter and Wheels (PharmaSlice*)	III	55. Rotary Cutter and Wheels (PharmaSlice*)	IIIIII	55. Rotary Cutter and Wheels (PharmaSlice*)
60. Heated Wire Gravity Slide	II	60. Heated Wire Gravity Slide	III	60. Heated Wire Gravity Slide
62. Scissors with Edge Clamps and Conveyor Belt	I	62. Scissors with Edge Clamps and Conveyor Belt		
69. Hinged Blade with Roller (SliceFlow*)	IIII	69. Hinged Blade with Roller (SliceFlow*)	III	69. Hinged Blade with Roller (SliceFlow*)

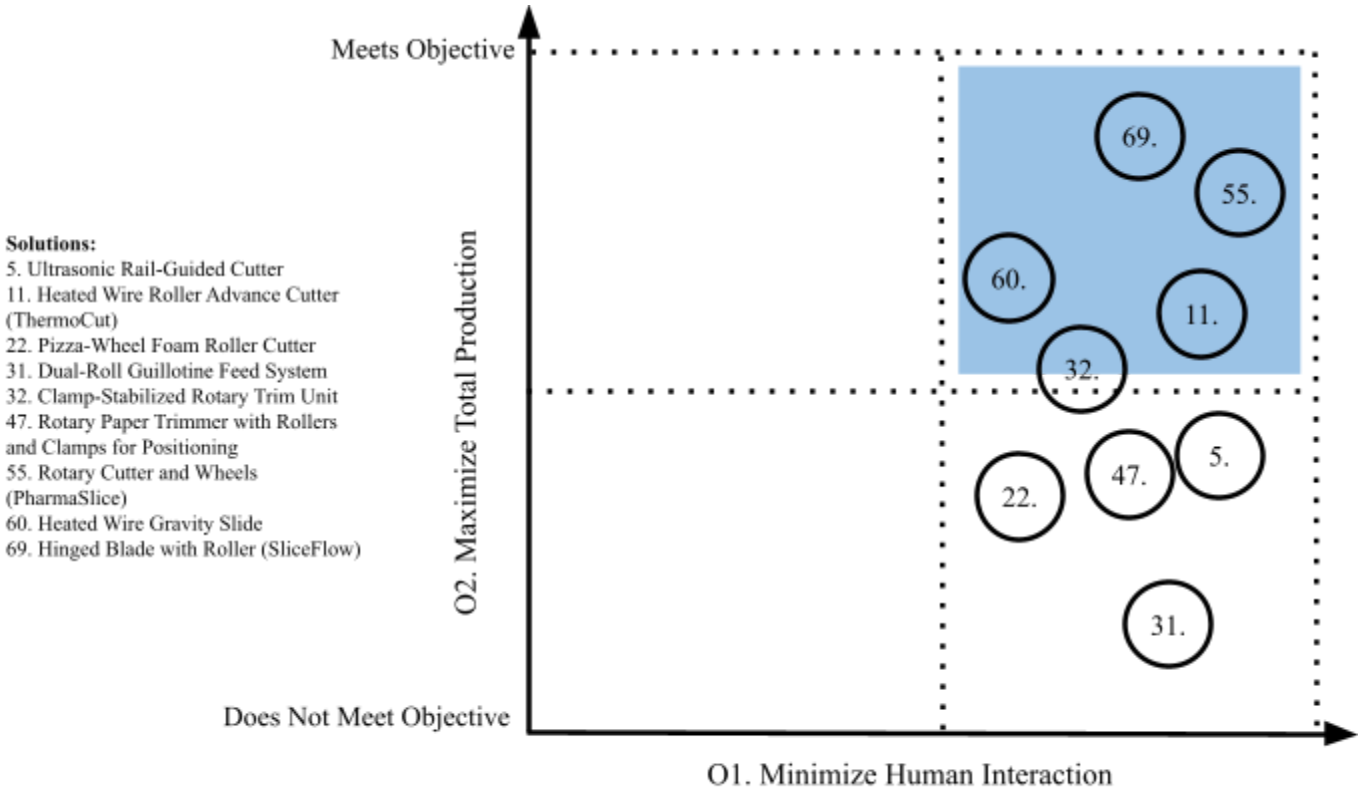
Final list after multivoting:

5. Ultrasonic Rail-Guided Cutter
11. Heated Wire Roller Advance Cutter (ThermoCut*)
22. Pizza-Wheel Foam Roller Cutter
31. Dual-Roll Guillotine Feed System
32. Clamp-Stabalized Rotary Trim Unit
47. Rotary Paper Trimmer with Rollers and Clamps for Positioning
55. Rotary Cutter and Wheels (PharmaSlice*)

60. Heated Wire Gravity Slide
69. Hinged Blade with Roller (SliceFlow*)

H.5: Graphical Design Selection

Section H5 presents the graphical comparison of the nine remaining ideas against the project's top two objectives. The three highest-scoring concepts then advanced to the Pugh Chart for the final evaluation step.



Appendix I: Pugh Method

Below is the pugh method used to compare each design against a datum (Design 1/PharmaSlice) across the seven project objectives, using a scoring scale from -2 to +2. Each design's relative performance was evaluated objectively, and the design with the highest overall score was selected as the final proposed solution.

Table I.1 The Pugh Chart

Objective	**Design 1 (PharmaSlice /Datum)**	**Design 2 (ThermoCut)**	**Design 3 (SliceFlow)**
O.1 Human Interaction	S	0	0
O.2 Total Production	S	0	0
O.3 Success Rate	S	0	0
O.4 Loading Capacity	S	0	0
O.5 Maintenance Frequency	S	+1	-1
O.6 Internal Visibility from Outside	S	-2	-1
O.7 Dimensions	S	-1	0
O.8 Spending	S	-1	0
Total	**0**	**-3**	**-2**

Justification:

O.1 Human Interaction:

- ThermoCut: As you have to set up the strip in between the two wheels, it will take longer to set up than the datum where only one pouch needs to be fed.

- SliceFlow: Follows the same loading mechanism as the datum therefore it has score 0.

O.2 Total Production

- This metric is unable to be objectively compared as it would need to be physically tested for both designs.

O.3 Success Rate

- This metric is unable to be objectively compared as it would need to be physically

tested for both designs.

O.4 Loading Capacity

- The loading capacity of all designs are limited by the length of the strip and therefore all the same.

O.5 Maintenance Frequency

- ThermoCut: As the design relies on heat to cut through the material, the blade will not dull over time and therefore will not need to be replaced as often.

- Slice Flow: As the design uses a flat blade it has less surface area and therefore degrades faster than a round blade.

O.6 Internal Visibility from Outside

- ThermoCut : As the cutting system is within a shell (to provide safety), there is limited visibility.

- Slice Flow: This design uses a single flat blade rather than a full 360 degree blade meaning that there is less cutting surface area and would therefore dull faster / need to be replaced more often.

O.7 Dimensions

- ThermoCut: H: 60 > 30.5cm , W: 40 > 30.5cm, L: 50> 30.5cm therefore it is larger than the datum.

- SliceFlow: H: > 30.5cm , W: 24 < 30.5cm, L: 40 > 30.5cm therefore the overall size is similar to the datum.

O.8 Spending

- ThermoCut: As this design requires a power supply to heat the wire, it will cost more than the datum.

- Slice Flow: Both designs will cost approximately the same amount as the main materials used are the same but the configuration differs.

Appendix J.1:

Justification for each objective analysis for design 1.

Objective	Analysis
Human Interaction	The design only requires the pharmacy technicians to load the strip into the machine's intake and collect the separated packages from the output basket, with occasional clearance of jams if they occur. The wheel advances packages automatically through the cutting zone, while the Arduino motor triggers the cutter at precise intervals to cut exactly 2,

	5 or 10 packages. Each strip contains anywhere from 100 to 500 pouches, at 8000 packages a day, which will require 16-80 actions a day. Taking a rough estimate of 30 seconds per load and calibration, the max loading time would be 40 minutes. Allowing 15 minutes for occasional jams, this design stays under the 60 minutes required of O.1.
Total Production	Production rate is determined by the roller advancement speed and the rotary cutter wheel speed. The roller system advances strips at a maximum of 12.57 cm/s (based on a standard 60 RPM motor with a 40mm drive pulley), advancing one pouch length (6 cm) in roughly 0.5 seconds. The rotary cutter wheel is also powered by the same motor, with a wheel diameter of 45mm, the circumference is 14.14cm. To cut across the 4 cm package width, the wheel requires approximately 0.28 seconds to complete one cutting pass. Total cut cycle time is approximately 0.28 seconds, with no pre-heating or cooling required. Detailed production specifications for each batch size are below: ● **2-package batches**: with a strip length of 12 cm, feed time is 1.0 second plus 0.28 second cut cycle, totaling 1.28 seconds per cycle. This yields 2813 strips per hour (5,625 pouches/hour), exceeding the 1750 strips/hour requirement and completing the 8000-pouch daily goal in 1.4 hours. ● **5-package batches**: with a strip length of 30 cm, feed time is 2.5 seconds plus 0.28 second cut cycle, totaling 2.78 seconds per cycle. This yields 1295 strips per hour (6475 pouches/hour), below the 1750 strips/hour requirement, completing the 8000-pouch daily goal in 1.2 hours. ● **10-package batches**: with a strip length of 60 cm, feed time is 5.0 seconds plus 0.28 second cut cycle, totaling 5.28 seconds per cycle. This yields 682 strips per hour (6818 pouches/hour), below the 1750 strips/hour requirement, completing the 8000-pouch daily goal in 1.2 hours.
Success Rate	Guide rails constrain the strip to a precise path, keeping the pouches aligned horizontally. Friction from the advancement wheels provides consistent advancement without slippage, ensuring each cut occurs at the perforation consistently. The Arduino can be calibrated to optimize cutting timing for different cutting intervals, allowing fine adjustments to reduce the risk of misaligned cuts. With these physical and technical constraints and the ability to calibrate, the design

	is projected to achieve at least a 99.9% success rate, exceeding the objective requirement.
Loading Capacity	The open feed path has no physical stop or length limitation; strips can be fed until the entire roll is processed. The drive wheels maintain consistent grip regardless of strip length, while the guide rails and friction rollers prevent bunching, even on long strips. This allows a full input roll of 500 packets, meeting objective requirements.
Maintenance Frequency	All components are available off the shelf and are easily replaceable without specialized tools. The rotary trimmer blade is the component with the most potential wear. Industrial-grade slitter blades used for lightweight cloth fibreglass last up to 20000m worth of cutting length, at an average thickness of 100 microns, similar to our material thickness. At 4000 cuts a day for 2-package batches, this extends the blade life to over 125 days (4 months). [36][37]. Brushless DC motors are rated for up to 20000 hours of operation, taking the average at 10000 hours, working for a maximum of 3 hours a day; this is over 9 years of operation. [38].
Internal Visibility from Outside	The open-frame construction leaves the entire cutting mechanism and strip path exposed to view; however, it could be optionally covered up during operation. The design would have a box around the cutting blade, protecting the operator from the exposed blade. This visibility allows for early detection of jams or clearance of jams.
Dimensions	The prototype would have a 305mm x 305mm footprint with its base. Well within the dimension objectives, allowing placement on any standard pharmacy countertop 630mm - 762mm [39]. This compact size also allows for easy relocation if the pharmacy layout changes.
Spending	The total cost of materials will be under $200 for the prototype, with the final design benchmarked using a similar vinyl cutter machine priced at $650 at the time of writing (March 20, 2026) [40]. This leaves over $9000 dollars to modify the machine or to specifically tailor it to our needs.

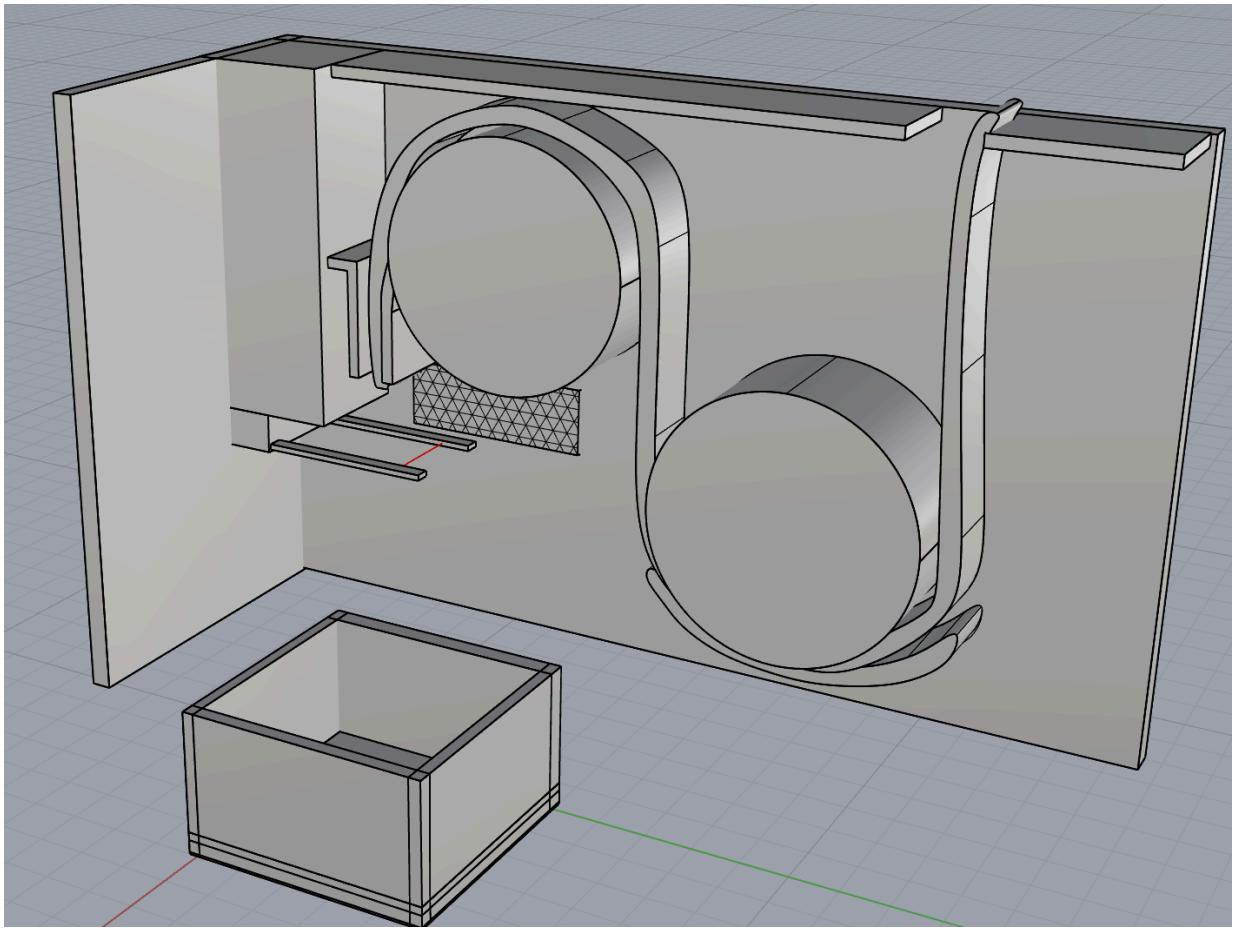
Appendix J.2:

Justification for each objective analysis for design 2, with a 3D rendering attached for clarity.

Objective	Analysis
Human Interaction	The ThermoCut design requires the operator to load strips into the machine and collect periodically from the collection basket. The DC motor-driven wheels automates strip advancement through the cutting zone, with a dual wheel design to ensure sufficient control over the packets, preventing slipping. The heated wire cycles at speeds corresponding to intervals of 2, 5, or 10 package lengths, determined from prototyping data. To avoid complexities arising from more wheels used in aligning the strips, each batch must be manually loaded and slotted between the wheels, as the heated wire with limited roller mechanism does not accommodate continuous roll feeding as seamlessly as the other two designs. Based on 8000 packages per day, at an average 100-500 packages a strip, the operator would load 16-80 packages per day. At 45 seconds per load, due to manual alignment, total loading time is 60 minutes. Adding in 20 minutes to collection and resolving occasional jams, this brings a max daily total to 80 minutes, exceeding the 60 minute objective target.
Total Production	Production rate is limited by the roller advancement speed and wire cycle times. The roller system advances strips at a maximum of 12.57 cm/s (based on a standard 60 RPM motor with a 40mm drive pulley), advances one pouch length (6 cm in length) in roughly 0.5 seconds. The heated wire cuts at a rate of 400mm/min, 6.7mm per second [25]. At a thickness of 100 microns, it takes 0.015 seconds to cut, and 1 second to cool down and heat up, for a total cycle time of approximately 1.015 seconds. Detailed production specifications for each batch size are listed below: ● **2-package batches**: With a strip length of 12 cm, feed time is 1.0 second plus 1.015 second cut cycle, totaling 2.015 seconds per cycle. This yields 1788 strips per hour (3576 pouches/hour), exceeding the 1750 strips/hour requirement and completing the 8000-pouch daily goal in 2.2 hours. ● **5-package batches**: With a strip length of 30 cm, feed time is 2.5 seconds plus 1.015 second cut cycle, totaling 3.515 seconds per cycle. This yields 1024 strips per hour (5120 pouches/hour), below the 1750 strips/hour requirement, completing the 8000-pouch daily goal in 1.6 hours. ● **10-package batches**: With a strip length of 60 cm, feed time is 5.0 seconds plus 1.015 second cut cycle,

	totaling 6.015 seconds per cycle. This yields 598 strips per hour (5980 pouches/hour), below the 1750 strips/hour requirement, completing the 8000-pouch daily goal in 1.3 hours.
Success Rate	Success rate is determined by the guide rails and precise temperature control. Internal guide rails constrains the strips to a path, maintaining alignment within ± 1mm of the cutting zone throughout advancement. The nichrome wire is heated to 300 degrees, 70 degrees above the melting point of PET, ensuring clean cuts through packaging without significant contact force [26]. This temperature margin accounts for heat dissipating to the surrounding material to ensure consistent cuts even with variations in plastic thickness. The heated wire melts through the perforated seam instead of cutting or tearing, reducing the risk of package damage. With these design features, the expected success rate is to be above 99.9%.
Loading Capacity	The roller intake has no physical stop or length restriction. Strips can be loaded end to end, though manual alignment is required for each new strip. The ATP machine's maximum output of 500 packets per roll can be loaded and processed continuously, meeting objective requirements.
Maintenance Frequency	The nichrome wire is the primary wear component in this design. Nichrome has a maximum operating temperature of 1050 degrees [27], well above the 300 degree service temperature, ensuring the wire remains within its limits and does not degrade rapidly. Under continuous operation at 300 degrees, significantly lower than the rated temperature, nichrome wire only requires replacement when it becomes oxidized, upwards of thousands of hours of operation [41]. The filter should be replaced every 3 to 12 months, as PET heated to 200-300°C is expected to release only acetaldehyde and formaldehyde while retaining high thermal stability till 382°C (< 0.4 wt.% loss) [42] [43], any hydrophilic zeolite filters selected for these aldehydes should be replaced conservatively based on prototype testing [44]
Internal Visibility from Outside	A 130mm x 180mm transparent glass panel is mounted on the side of the enclosure, providing a viewing window into the cutting zone. This window allows operators to see the strip alignment and the wire's operations. Operators can see only the critical cutting area but not the entire strip path due

	to the complex internals. The glass panel is heated and coated to prevent fogging to vaporized plastics.
Dimensions	The ThermoCut design integrates all components into a compact footprint, within 305mm x 250mm, and a height of 250mm. This is well within the dimension objectives, allowing placement on any standard pharmacy countertop 630mm - 762mm [45]. This compact size also allows for easy relocation if the pharmacy layout changes.
Spending	Nichrome wires, hydrophilic zeolite filters and small motors are easily accessible at low cost on online platforms. [46] Using aluminum as a benchmark casing material (except ceramic casing around wire for thermal protection), the prototype is expected to cost less than $600 CAD as of March 20, 2026 with materials, components, and processing accounted for. [47]



Appendix J.3:

Justification for each objective analysis for design 3.

Objective	Analysis
Human Interaction	The SliceFlow design requires the pharmacy technicians to only periodically load the strip and collect finished strips. The dual-conveyor system automates advancement through the cutting zone, while the motor-controlled blades cycle automatically at intervals corresponding to 2, 5, or 10 package lengths. The cleated slider and foam roller ensure consistent package and pill positioning without manual adjustment. Based on 8000 packages a day, and an average of 100 to 500 packages a strip, the operator loads 16 to 80 times a day. At an upper limit of 30 seconds per load, total loading time maxes out at 40 minutes. The foam roller and conveyor system operates continuously without intervention, and output packages collect in a bin, requiring collection only after needed or filled. Clearing occasional jams, estimated to be 10 minutes daily, brings the total interaction time to 50 minutes, meeting the objective goal of $\leq$ 60 minutes.
Total Production	Production rate is determined by conveyor advancement speed and blade cycle time. The conveyor belt system advances strips at a max of 12.57 cm/s (based on a standard 60 RPM motor with a 40mm drive pulley), positioning each cutting strip under 1 second each. The motor-controlled blade descends for 0.2 seconds, cuts for 0.1, and retracts for 0.2, for a total cut cycle of 0.5 seconds. The dual-conveyor design allows the next package to advance while the blade retracts, allowing for the next package to advance while the blade retracts, overlapping interactions. Each cycle, with the full advance and cutting sequence, timing expectations are below, using 6 cm for each pouch length: ● **2-package batches**: With a strip length of 12 cm, the feed takes 1 s plus a 0.5 s cut cycle; set at 1.5 seconds per cycle, this yields 2400 strips/hour (4800 pouches/hour), exceeding requirements and completing the 8000-pouch goal in 1.7 hours. ● **5-package batches**: With a strip length of 30 cm, the feed takes 2.5 s plus a 0.5 s cut cycle; set at 3 seconds per cycle, this yields 1200 strips/hour (6000 pouches/hour), below requirements, but still completing the 8000-pouch daily goal faster than the 2-package batches, in 1.3 hours. ● **10-package batches**: With a strip length of 60 cm,

	the feed takes 5 s plus a 0.5 s cut cycle; set at 5.5 seconds per cycle, this yields 654 strips/hour (6545 pouches/hour), below requirements, but still completing the 8000-pouch daily goal faster than the 2-package batches, in 1.2 hours.
Success Rate	Success rate is controlled by the pill repositioning system and package alignment. Before reaching the cutting station, the foam roller rotates to gently sweep each pill towards the side of its package. The roller's foam compresses slightly (3-5mm) to accommodate pill thickness while applying just enough force to reposition without crushing. This ensures pills are clear of the blade path. The cleated slider path uses low ridges that guide the edges of the pouches, maintaining alignment within ±0.5mm from left to right shifting. The conveyor gap (20 mm) allows the blade to pass completely through the package without contacting the conveyor surface, ensuring clean cuts. The acrylic cover allows for visibility while protecting the operator against cutting fragments. With these design features, the theoretical maximum failure rate from misalignment is low, with misaligned blades even lower, suggesting a high success rate exceeding that of the objective.
Loading Capacity	The conveyor belt intake is open with no physical stop or length restriction. Strips can be loaded end-to-end or fed from a roll. The conveyor belts (minimum width 42mm) and cleated slider bed maintain consistent traction regardless of slip length, preventing bunching or slipping during long runs. A full 500 pouch roll can be loaded at once and processed without operator intervention, meeting objective requirements.
Maintenance Frequency	As with design 1, SliceFLow will use a DC motor to power the cutting mechanism. These motors are rated for up to 20000 hours (9 years) of operation before needing replacement [48]. The limiting factor will be the dulling of the blade with wear. The average lifespan of blades used for cutting paper and plastic films is 80,000–100,000 cuts before noticeable wear. With a maximum 8,000 pouches (or 4,000 cuts) per day, each blade would last around 3 weeks before any noticeable wear [49]. However, it would take nearly 6-12 months before there are any critical drops in performance [50].

Internal Visibility from Outside	Plexiglass was chosen for its high optical clarity, strength and stiffness [51] in order to ensure staff safety while preserving full visibility of internal mechanical components and operations. The removable acrylic cover provides 360-degree visibility of the entire operation. Unlike designs with limited viewing windows, the SliceFlow's transparent enclosure allows the operator to see all internal mechanical parts. This exceeds the 180 mm visibility requirement substantially. The clear acrylic material retains over 90% of the light passed through it, allowing for little distortion when viewing. This allows for early detection of jams or misaligned packages, leading to quick intervention [52].
Dimensions	The SliceFlow design footprint is determined by the conveyor belt system; total length is 40 cm, including the input area, the cutting station, and the output chute. Total width is expected to be around 24cm, allowing for mechanical parts to fit within the frame. The complete unit dimensions are 40 cm L x 24 cm W x 15 cm H, well within the objective limits of 80 cm by 62 cm. This compact footprint allows placement on any standard pharmacy countertop [45].
Spending	The total cost of materials will be well under $10000 CAD. Benchmarking off of a tabletop wood saw, which has a more expensive cutting mechanism, but lacks the conveyor mechanism, costs around $500 CAD as of March 20, 2026 [53].

Appendix K:

Additional information and justifications for the methods of success.

Appendix K.1: Selected Objective and Justification

The team's top critical objectives for validating design success, ranked through pairwise comparison (Appendix E) and client feedback [CLIENT SOURCE], are detailed below:

- O.1: Minimize Human Interaction With Design (< 60 min/day)
- O.2: Maximize total production (≥1750 strips/hour)
- O.3: Maximize success rate (≥ 99.9% undamaged packages)

Three objectives are functionally dependent on each other in the proposed design and in alignment with the client's requests. Testing them separately would fail to capture the design's real-world performance requirements.

These interdependent design objectives tie back to each function. The design must advance the strip precisely (connects to O.2), secure packages without crushing pills (O.3) and cleanly cut at the perforation line (O.2, O.3). Failures are directly linked to all three objectives:

- **Opened packages due to misaligned cuts** (O.3 failure) require human intervention to remove the damaged medication and reset the machine, increasing human interaction time (O.1) and reducing production (O.2)
- **Misalignment or feeding errors** (O.3 failure) create inconsistent cut quality, damaging the quality of medication, resulting in waste and additional human interactions (O.1).
- **Mechanical jams from slippage or bunching** (O.2 failure) stop production completely, forcing technicians to manually clear the machine, increasing human interactions (O.1) and potentially damaging packages (O.3)

The client's main problem with repetitive strain injuries is addressed by O.1; however, O.2 and O.3 are prerequisites. A design that doesn't meet the production quotas or damage packages will not be used; therefore, the design must compare to all three objectives to ensure it is practical for pharmacy use.

Appendix K.2: Test Method Overview

The test will employ bench-top testing using the team's functional prototype of the rotary paper trimmer design. This method was selected as it allows for direct measurement of all three interdependent objectives under simulated working conditions, using hardware that would be later used in the pharmacy. [32]

Bench-top testing is a standard validation approach used in medical devices and biomedical engineering, where products are tested on a small scale in a controlled setting to verify the key mechanisms and functional performance before being deployed. This method is essential in ensuring the design meets our testing objectives to identify key issues early on. For our prototype, bench-top testing is appropriate, as it allows for direct assessment of cutting precision, speed, and required maintenance. [32][33]

Prototype Configuration:

- Rotary paper trimmer mounted on a wooden rail
- Wheel-driven advancement of the strip along a guided path for precise positioning
- Computer-controlled cutting intervals during testing (final design will use buttons or dial)
- Computer control ensures consistency between tests, allowing the team to focus on evaluating performance objectives

Test Personnel:

- Two team members will conduct testing:
    a. Member 1: Manages machine operation (loading strips, clearing jams)
    b. Member 2: Tracks time and records data
- A two-person team allows for precise documentation while maintaining consistent and safe operations [34]

Test Materials:

- Authentic hospital medical strips (~200 packages) provided by the client
- Office letter paper, roughly 110-112µm [https://action-press.co.uk/blog/paper-thickness-and-weight-explained/]
    - LDPE film: 38 ± 2µm
    - PET layer: 12 ± 2µm
    - Total one-sided thickness: 110-118µm (verified per ASTM D6988)(Sinai Health Pharmacy, private communication, Feb. 2, 2026)

According to the plastic specifications provided by the client, the pouches are made from LDPE and PET with a combined thickness of 110-118 µm (both layers) (Sinai Health Pharmacy, private communication, Feb. 2, 2026). According to Health Canada (GUI-0029), medical device validation shall be conducted using the actual material the device will encounter or other highly similar samples when authentic samples are limited. [35].

Appendix K.3: Test Equipment, Procedure and Data Collection

Prototype Components:

- Wooden frame prototype
- 2 DC gear motors (12V, 60RPM)
    - Motor 1: drives the paper trimmer cart along the rail
    - Motor 2: Drives the wheel to advance strips
- L298N 2A motor driver
- Arduino UNO R3
- Breadboard
- Jumper wires (M-M, M-F)
- 22 AWG solid core wire
- Plywood sheets (¼” thick, 12”x12”)
- EVA black foam (½” thick, 12”x12”)

Testing Materials:

- Authentic hospital medication strips (~200 provided)
- Standard office paper (~40 sheets)

Measurement and Data Collection:

- A smartphone for video and a stopwatch
- Digital table for data collection

Output:

- Collection basket for separated packages

The prototype will be constructed on-site, using pre-purchased materials and on-site tools and machinery.

Test preparation:

1. Cut paper into 5 equal-width strips, connect strips lengthwise with tape, and prepare roughly 50 pouch lengths per strip, 10 strips total.
2. Use leftover material to run calibration tests:
    a. Verify motor 2 advances the correct distance, no slipping
    b. Verify motor 1 drives the cart trimmer smoothly

The calibration process ensures consistent and repeatable cutting distances before data collection begins, allowing the team to pinpoint key issues if they arise.

Test Execution (9 runs total, 3 for each cutting interval, twice with paper, once with strips)

1. Load the strip into the machine, ensure drive wheel is properly engaged with strip
2. Start the stopwatch and machine together
    a. Record timestamps of irregularities
3. If jam occurs, stop the machine, identify flaw, fix flaw and resume testing
4. At end of run, stop stopwatch, and calculate the number of packages processed per unit time
5. Check for any misaligned cuts, torn packages, or calibration issues in the long run
6. Repeat for 9 runs

Table K.1 details the data collected during each run and the measurement method

Table K.1: Data to be collected per run

Objective	Data collected	Measurement Method
O.1: Human Interaction	Loading time per strip	Video timestamp
	Intervention time per jam	Video timestamp
	Total run interaction time	Sum of all action times

O.2: Production Rate	Total run time per strip	Video timestamp
	$Package\ per\ minute\ = \frac{Total\ Packages}{Total\ runtime}$	Calculated
	$Proj.\ hourly\ rate\ =\ Pkg\ per\ min\ *\ 60$	Calculated
O.3: Success Rate	Total packages processed	Count
	Number of failures	Visual inspection + video
	$Success\ rate\ =\ 100\%\ *\ \lvert\frac{total-failure}{total}\rvert$	Calculated post-run

Data collection table example: screenshots are taken; however, the file is linked here
TUT#0135_Team181_CDS_Data_Collection_Template.xlsx

Master Data Log:

Run #	Date	Operators	Strip Material	Start Time	End Time	Total Pkgs	Total Run Time (min)	Total Int Time (sec)	FC-1	FC-2	FC-3	FC-4	FC-5	FC-6	FC-7	FC-8	Total Failures	Notes	Success Rate (%)	Pkgs per Minute	Hourly Rate	Daily Interaction (min)
EXAMPLE	2026-03-23	James/ Alba	Paper	10:00	10:04:30	120	4.5	45	0	0	0	0	1	0	0	0	1	Minor jam at roller, cleared quickly	99.17	26.7	1600	50.0

Event Log:

Run #	Timestamp (mm:ss)	Failure Code	Description	Duration (sec)
EXAMPLE	1:23	FC-5	Strip jammed at entry roller	15
EXAMPLE	3:47	FC-3	Cut 3mm off perforation line	0
EXAMPLE	5:12	FC-1	Pill crushed in package #42	0

Appendix K.4: Test schedule

The test will be conducted over three weeks following the CDS submission. Table K.2 details the tasks required for each week.

Table K.2: Three-week-long testing schedule

Date	Task	Personnel and location
Week 1 (Mar 23-29)	Prototype assembly, Arduino code, motor calibration, strip preparation	All team members at Myhal MyFab

Week 2 (Mar 30-Apr 5)	Calibration runs, tune if needed, testing runs.	Operator/Recorder at Myhal MyFab
Week 3 (Apr 6-8)	Complete data analysis, review video footage, compile final results	Operator/Recorder at Myhal MyFab

Appendix K.5: Success Criteria

The proposed rotary paper trimmer design will be considered successful if testing demonstrates that all three interdependent objectives are met together. Table K.3 details the success criteria for each objective.

Table K.3: Success Criteria by Objective

Objective	Metric	Target	Measurement Method
O.1 Human Interaction	Minutes spent interacting with the machine per day	< 60 minutes total (loading + interventions)	Sum of all loading times and intervention times recorded via stopwatch; projected to 8000 packages/day
O.2 Production Rate	Strips cut per hour	≥ 1750 strips per hour	Calculated from total run time and total packages processed; adjusted for cut interval
O.3 Success Rate	Undamaged packages per 1000 packages	≥ 99.9%	Visual inspection of all packages; failures categorized as misaligned cuts, torn packages, or jams

If the prototype barely misses one criterion but exceeds in others, the team will make note of this gap and make design modifications or future modifications for the design.

Appendix K.6: Limitations and Assumptions

The following limitations and assumptions apply to this test plan.

Limitations:

1. Limited samples: Only ~200 samples were available for testing; the majority of data will be collected using similar materials, which may not perfectly replicate the properties of the pouches
2. Visual inspection constraints: Without magnification tools or scales, minor damages to packages may go undetected, overstating the success rate

3. Controlled environment: Testing will be done in a university workspace, rather than an active hospital pharmacy. Results may not account for real-world interruptions or integration challenges.
4. Limited run count: With only 9 total runs, the sample size may not be large enough to capture all potential failure modes in performance

Assumptions:

1. Simulated paper strips (110-112 μm thickness) are reasonably close to the actual thickness, as both materials fall within similar thickness ranges
2. Results from approximately 1000 test packages can be reasonably extrapolated to a full 8000 packages
3. The two-person testing operation provides more reliable data, as tasks are distributed strictly
4. Visual inspection is sufficient to detect failures that would compromise medication integrity in a real pharmacy setting